

TPS System Planning Guide
For Windows 2003/XP

TP10X-400

TPS R400

7/2008

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About This Document

This document provides guidelines for planning and defining a TotalPlant Solution (TPS) system. It includes the system level planning, network planning, security, and performance considerations.

Release Information

Document Title	Document ID	Release Number	Publication Date
<i>TPS System Planning Guide</i>	TP10X-400	TPS R400	7/2008

References

Document Title	Document ID
<i>Experion HMIWeb Display Building Guide</i>	EP-DSX164
<i>Experion Network and Security Planning Guide</i>	EP-DSX174
<i>Experion Server and Client Configuration Guide</i>	EP-DSXX24
<i>Experion Server and Client Planning Guide</i>	EP-DSX134
<i>GUS Display Authoring Tutorial</i>	GU07
<i>HCI Client Developer's Guide</i>	TP60
<i>Integrated Experion-TPS Nodes (ES-T, ESVT, and ACE-T) User's Guide</i>	EP-DCX494
<i>(Hardware) Planning, Installation and Service Guide</i>	Refer to the Honeywell guide for your hardware platform.
<i>PHD Security and Network Planning Guide</i>	AM065 (in the PHD book set)
<i>System Management Configuration Guide</i>	EX08 -300 or EP-DSX204
<i>TPN Engineer's Reference Manual</i>	SW09-605
<i>TPS System Administration Guide</i>	TP06x-400
<i>TPS System Implementation Guide</i>	TP08x-400

Contacts

World Wide Web

The following Honeywell web sites may be of interest to Process Solutions customers.

Honeywell Organization	WWW Address (URL)
Corporate	<i>http://www.honeywell.com</i>
Honeywell Process Solutions	<i>http://www.honeywell.com/ps</i>

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the Honeywell bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the Honeywell bookset.

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1. Introduction

1.1 About this Document

Purpose

This document provides basic guidelines to plan and define a Honeywell TotalPlant Solution (TPS) system. It includes the system level planning, network planning, security planning, and performance considerations.

Use this document in conjunction with the TPS Network (TPN) Planning Guide and other TPS application-specific planning topics, such as TPS Builder.

Users of this document should be familiar with Windows 2003/XP administration and planning, and have network configuration skills.

Releases relevant to this document

This document is relevant to the following releases of TPS system software (where “xx” represents a series in the software release, such as GUS R400):

- APP R4xx
- GUS R4xx
- Experion R3xx
- PHD R2xx
- TPN R53x and R6xx
- TPS (Infrastructure) R4xx
- TPS Builder R4xx

1 Introduction

1.1 About this Document

2. System Planning and Analysis

2.1 Overview of Planning and Analysis Tasks

Concepts you should know

This section provides information and guidance in the following areas along with specific Honeywell recommendations:

- TPS Node functionality
- LCN-connected Experion nodes
- Sample TPS system topologies
- Industrial Strength Networking configuration and components

The rationale and the functionality are provided for each architecture. The guidelines and principles are intended to provide a TPS Administrator with an understanding of the TPS system dynamics to facilitate the construction of more complex arrangements of TPS nodes when required.

Segregate Process Control Network from Plant Control Network (PCN)

In general, the overall system partitioning approach is to segregate the dedicated process control system from the "open" plant information network world, composed entirely of commercially available platforms and networks. The TPS system consists of a standard set of open and proprietary network pieces, along with a minimum set of services to make it operational.

Guidelines based on standard configuration

If you choose to change this standard configuration, guidelines are available to assist in doing so, but performance benchmarks, TAC support, documentation, and testing are oriented around the "standard" implementation. Therefore, you are encouraged not to change the standard configuration. However, if you decide you must change it, we suggest that you use Honeywell Support Services to help you.

2 System Planning and Analysis

2.1 Overview of Planning and Analysis Tasks

Key considerations for most topologies

Some of the key considerations that apply to most topologies are discussed next. This includes:

- Basic TPS Node Configuration and Guidelines
- Basic Network Configuration and Components
- Basic Windows 2003 Domain Configuration Guidelines
- TPS Domain Implementation

Resource Kit planning tool

The Windows 2003 Resource Kit provides a valuable source of tools and documentation for resource planning, networking, system administration, and diagnostics. Consider purchasing the resource kit or download a subset of the capabilities from the Microsoft web site.

Event filtering through the MMC snap-in

Honeywell provides a Microsoft Management Console (MMC) snap-in that allows you to create a table for any event registered on the system including Windows XP/2003 and TPS system events. For more information refer to the System Management Configuration Guide.

2.2 TPS Node Planning

Human interface nodes

The number of human interfaces and displays available to the operator needs to be considered and planned. In general, these stations have little third-party content and are focused at achieving the operator's control and monitoring objectives. They are typically connected to the TPN for accessing data quickly, but may also retrieve data from external sources such as Process History Database (PHD) or an application suite.

Human interface nodes include Experion Station – TPS (ES-T), Experion Station – Flex (ES-F), Global User Station (GUS), and Universal Station (US).

Application nodes

The Application Processing Platform (APP) uses the TPS node hardware. As such, it contains both TPN and PC capabilities. The personal computer side of the node is a Windows operating system with a communication infrastructure consisting of Object Linking and Embedding (OLE), OLE for Process Control (OPC), and Distributed Component Object Modeling (DCOM). These are all Microsoft standard technologies for communication in an object-oriented environment. Also included is the Honeywell Communication Interface (HCI), which offers additional features beyond what is recommended by the OPC standards committee. Together these pieces are extended to all TPS nodes and applications to provide a common, open mechanism for communicating between nodes. Advanced control applications, information management applications, and third-party application software will execute in this PC environment.

Application node considerations

Evaluate the following considerations when implementing an application node:

Table 1 Application Node Considerations

Node Type Connectivity	Considerations
TPN-Connected (APP)	• The location of their data sources • The load that these nodes put on the PCN and TPN and their need for fast TPN data access • See Time Sync and Channel Configuration Considerations
Non-TPN-Connected	• Their data access needs and data sources from which they request data • The load that these nodes put on the PCN and the HCI/OPC servers • The need for any off-the-shelf large computing resources

Redirection Manager supports HCI/OPC server availability

HCI/OPC servers, such as TPN Server, as well as third-party OPC servers, can achieve high availability through the use of Redirection Manager. Redirection Manager monitors primary and secondary HCI or OPC servers for availability. Failures are detected quickly and communications (OPC calls) are transparently redirected to an available server.

Engineering stations

In the context of this document, an engineering station is any of the following:

- Experion Station – TPS
- Experion Station – Flex
- GUS node equipped with a write-enabled storage media, DVD/CD-ROM, and File Transfer

The following table details several engineering station components and their function:

Table 2 Engineering Station Components and Function

Component	Function
DVD/CD-ROM	Enables TPS system software to be loaded locally.
Write enabled storage media (CD/RW, DVD, Zip)	Enables local backup of TPS system files, including schematics, checkpoints, and Network Configuration File (NCF) to removable media.
File Transfer	Mechanism to move .EB and .CL files between a PC and the system. The standard file management utility, Windows Explorer, or a batch file can be used to move display files between an engineer's desktop and the domain controllers, or to individual workstations.
SafeView	Manages the display environment so operators do not lose their view of the process or alarm status.
TPS Builder Database	Can be built at the engineer's desktop using TPS Builder or stored on a server. Refer to TPS Builder documentation for guidelines.
HMIWeb Display Builder	Web-based display builder for Experion stations.
Local file storage	For picture files and TPN node reload purposes.



ATTENTION

Honeywell recommends that you configure at least one TPS node (Experion Station – TPS or GUS) on each TPN for use as an Engineering station with access to the Native Window application. This facilitates “desk-to-system” engineering and effective media backup.

Experion Server-TPS (ESVT)

The Experion Server-TPS provides all of the functionality of the Experion Server, plus additional hardware and software components specific to TPS. Some of the features and functions are:

- LCNP for direct connection to the TPN.
- FTE for connection to Experion controllers.
- Available in redundant configuration.
- Repository services for Experion components.
- Process and event history.
- OPC Server for data and events for advanced control and optimization applications.
- Host services for Experion Station – TPS (ES-T) operator stations (clients).
- Distributed System Architecture (DSA) that enables seamless integration of multiple Experion servers (including Experion Server –TPS). This integration includes data, alarms, history, trends, and reports.

Application Control Environment-TPS (ACE-T)

When an Experion Application Control Environment (ACE) becomes LCN-connected, the platform for the ACE is the standard ACE platform, with the addition of a LCNP card and selected components of the APP version of the TPS infrastructure. An Application Control Environment -TPS node provides all ACE functions, with no extensions. An LCN-connected ACE is not intended to serve as an APP. That is, non-ACE applications will not be supported on the ACE. Furthermore, ACE-T does not support AM point processing through its LCNP.

An Application Control Environment - TPS has the same relationship with TPN data owners as an ACE. On an Experion system, an Application Control Environment - TPS can participate in peer-to-peer relationships with other Experion controllers. On a TPN system, an Application Control Environment - TPS functions as an HCI Data Access client of TPN data owners, using its local TPN Server and LCNP. TPN data owners have no access to ACE data or functions.

PHD nodes

A PHD node collects, stores, and replays historical plant data, including continuous process data, material movements, operating events, quality data, and other types. It incorporates two types of databases:

- Real-time – for storage of continuous process history data, including both current real-time and history – referred to as the Process History Database.
- Relational – uses the Oracle RDBMS for storage of all other data including events/transactions and plant reference data.

A key PHD Server feature is its ability to shadow data to other PHD Servers on the Plant Information Network (PIN). This provides a common place to archive history from multiple control rooms. In addition, it allows users of the PIN to access process data without needing access to the PCN.



ATTENTION

Honeywell recommends that you configure at least one PHD node for each TPN. Consider using a shadowed PHD Server on the PIN for plant-wide data collection, storage, and management.

TPN interface channels

TPN interface channels are the resource for accessing data, alarms, and files from the TPN, as well as triggering Windows 2003/XP applications. A maximum number of these channels can be allocated on any given TPS node connected to the TPN. The following table gives the ranges and recommended defaults for the software components that use TPN interface channels (these are configurable for some components).

Only software components that have ranges are configurable.

Ensure that the maximum channels are not exceeded on any given node.

2 System Planning and Analysis

2.2 TPS Node Planning

Table 3 TPN Interface Channel Utilization

Software Component	TPN Interface Channels (32 Max)			
	Range		Recommended	
	APP	GUS	APP	GUS
Reserved	3	3	3	3
TPN Server ^{See Note 1}	1-29	1	10	1
CL/NT Application Server	2	NA	2	NA
Synchronize Windows 2003/XP time using TPN time	1	1	1	1
File Transfer	3	3	3	3
(HOPC)	NA	13	NA	13
TPN RDI for PHD ^{See Note 2}	1-29 1 per RDI	NA	2 1 per RDI	NA
TPN Journaling to PHD	1	NA	1	NA
GUS LCN File Load	NA	2	NA	2
Faceplates/Group Display ^{See Note 3}	NA	8	NA	8
GUS Alarm Summary	NA	1	NA	1
GUS Remote Displays Server ^{See Note 4}	NA	20	NA	20
Other TPS System Applications ^{See Note 5}	see appl. spec		see appl. spec	
NOTE 1: Only 1 channel is allowed on a GUS node to preserve GUS Data Server performance. NOTE 2: Each TPN PHD RDI reserves 1 channel. There is no upper limit on the number of TPN RDIs, but typically, a user will use 2 TPN PHD RDIs (one per collection rate). NOTE 3: Each GUS faceplate uses 2 channels. Each GUS can have up to 4 faceplates running. A GUS Group Display uses 2 channels. If a GUS Group display is running, then only 3 faceplates will be allowed. These are included in HOPC. NOTE 4: GUS Remote Displays Server reserves 4 channels for its internal use. Additionally, each remote client consumes 4 channels. The maximum number of remote clients is 4. If a node runs the GUS Remote Displays Server, then channel requests from GUS components (such as GUS Data Server, GUS Faceplates, Alarm Summary, and Group displays) get routed through the GUS Remote Displays Server. Therefore, the maximum of 20 channels covers all. NOTE 5: This includes applications such as RMPCT, TotalPlant Batch, and OM&S.				

Table 4 TPN Interface Channel Utilization for LCN-Connected Experion nodes

Software Component	Range	
	TPN Interface Channels (32 Max)	
	ESVT	ES-T
Reserved	3	3
TPN Server ^{See Note 1}	1-29	1-15
CL/NT Application Server	NA	NA
Synchronize Windows 2003/XP time using TPN time	NA	NA
File Transfer ^{See Note 2}	3	3
HOPC ^{See Note 3}	NA	14
TPN RDI for PHD	NA	NA
TPN Journaling to PHD	NA	NA
GUS LCN File Service	NA	NA
Classic TPN Faceplates/Group Display ^{See Note 4}	NA	NA
GUS Alarm Summary	NA	NA
GUS Remote Displays Server ^{See Note 5}	NA	17
Other TPS System Applications	NA	NA
NOTE 1: Eleven channels are recommended. Out of the 11 channels, 1 channel is fixed and the other channels are configurable (installation default = 10). NOTE 2: Optional application. NOTE 3: 0 channels if GUS Remote Displays Server is enabled NOTE 4: These are included in HOPC and GUS Remote Displays Server totals. NOTE 5: GUS Remote Displays Server is available on Experion Station – TPS (ES-T) only. It has a maximum of 17 channels and depends on the number of external GUS Remote Display clients connected per formula: Total = 8 + n*3 (n = 0, 1, 2, 3), where n = number of external clients. Example Channel Calculation: An Experion Station – TPS (ES-T) supports GUS Remote Displays and 2 GUS Remote Display clients and File Transfer. It will use 31 channels based on ((8 + 2x3 remote displays) + 11 recommended for TPN Server + 3 File Transfer + 3 Reserved).		

TPN Time Synchronization

With GUS R330 (or later) and APP R211 (or later) and Experion for TPS, the TPN Time Synchronization function provides the ability to synchronize the TPN time to the TPS node time. This function is not applicable to the TPS Universal GUS nodes.

If you plan to implement the TPN Time Synchronization function, the CLOCK NODE 1 and CLOCK NODE 2 clock source nodes, as defined in the Network Configuration File (NCF.CF), must have LCNP4-GPS boards (or later) installed. Assign CLOCK NODE 1 and CLOCK NODE 2 to the appropriate GUS and/or APP node numbers when configuring the NCF to enable the implementation of the TPN Time Synchronization function.

In addition, all TPN nodes must have the K2LCN board (or later) installed.

2.3 PCN Network Configuration and Components

Configuration introduction

The basic PCN network configuration relies on several components including cabling and connectivity equipment.

Cabling

The primary determination for PCN cabling is the desired throughput as detailed in the following table:

Table 5 Network Cabling Requirements

Desired Throughput	Minimum Required Cabling	Maximum Segment Length
10 Mbps	Cat 3 UTP	100 Meters (328 ft.)
100 Mbps	Cat 5 UTP	100 Meters (328 ft.)



ATTENTION

Honeywell recommends utilization of Category 5 or 5e (fiber optic) cabling on the PCN to fully support current and future technology.

Ethernet switch

A switch provides the basic node-to-node connectivity. Additional capabilities are as follows:

- Provides a convenient means to connect multiple TPS stations.
- Enables the process control administrator to control connections on the workstation-side of the switch without the need for IT involvement.
- Provides network traffic isolation.
- Increases available bandwidth by providing dedicated bandwidth for each port.



ATTENTION

Honeywell recommends using a switch that supports 100 Mbps communication on each port.

Router

A router provides connectivity, security, and traffic isolation between LANs. A router is recommended when communication from the Plant Intranet to the Plant Control Network is required. The following table shows basic router capabilities:

Table 6 Router Capabilities

Item	Description
PIN Traffic Isolation	PIN network traffic is contained within the PIN segment. For example, the high network loading frequently experienced at the start of the working day on Plant Information Networks as users access e-mail, and so forth, are not imposed on the PCN. PCN Reliability is increased as a result of PIN network anomaly containment.
PCN Traffic Isolation	Local PCN network traffic is contained within the PCN segment and is not broadcast across the PIN. This minimizes PIN traffic loading.
Security	A router provides minimal security through the use of IP address filtering. Stronger security measures may include use of Virtual Private Networks (VPN) and/or a firewall. Note: Port Filtering and other security implementations may limit or prohibit certain functions such as DCOM callbacks.



ATTENTION

Honeywell recommends using router and IP address filtering when Plant Information Network to Plant Control Network connectivity is required.

Network communication protocols

TCP/IP is required for TPS system communications between nodes. Use of other protocols within a TPS node may impact security and system performance; however, other protocols can exist on the subnet subject to maximum loading.



ATTENTION

Honeywell recommends that TCP/IP should be the only protocol communicating on the PCN.

Honeywell Fault Tolerant Ethernet option

The Honeywell Fault Tolerant Ethernet (FTE) represents a way to achieve Ethernet redundancy through the use of Honeywell's FTE driver and redundant commercially available equipment.

Fault Tolerant Ethernet components allow network communication to occur over a functioning route. If that route should fail and another route exists, then communication occurs over that route.

Conventional Ethernet redundancy schemes typically employ two Ethernet networks, with each node (server or station) connected to both. If there is a failure on one network, the time for a node to switch to the other network can range from under 10 to over 30 seconds, depending upon network complexity and the equipment used. FTE, however, employs a single network and does not require a node to re-establish its network connection. As a result, switchover time is much faster—less than one second.

By providing twice as many communication paths between nodes, FTE also tolerates more faults—all single faults and many multiple faults. In addition, FTE is transparent to higher-level applications, which benefit from the high network availability FTE provides without requiring additional software configuration. Normal Ethernet nodes (non-FTE) can also connect to an FTE network and benefit from a more available communications environment than conventional Ethernet redundancy schemes offer.

Refer to the following documents for detailed information about FTE:

Doc ID	Document Title	Description
FE02	<i>Fault Tolerant Ethernet Overview and Planning Guide</i>	Contains theory and overview information for FTE networks, including Honeywell recommendations for network architecture. Also has switch configuration procedures.
FE05	<i>Fault Tolerant Ethernet Installation and Service Guide</i>	Contains installation and troubleshooting information for FTE nodes. Includes step-by-step procedures for installing FTE software on TPS or Experion nodes.

Experion Server redundancy

Experion servers, including Experion Server TPS (ESVT) Server, support redundant server configurations that provide warm fail-over architecture with online database replication. This differs dramatically from the more typical PC-based systems, which run two independent servers that are **not** synchronized at the database level. The Server redundancy scheme supports temporary removal of a server for maintenance. When the server is brought back on line, the databases can be re-synchronized at the touch of a button.

Multicast communication supports HCI name resolution

Multicast communication allows methods to resolve HCI alias names using an HCI Name Service. Name resolution can be accomplished in one of two ways:

- Multicast
- HCI alias file lookup

If multicast is used, networking equipment should be configured not to pass multicast messages from the PCN to the PIN. Refer to the System Management Configuration Guide for a discussion of setting multicast groups. Refer to the FTE Planning, Installation and Service Guide for information about disabling IGMP snooping on switches.



ATTENTION

Honeywell recommends not to pass PCN multicast messages to the PIN. Additionally, to minimize the opportunity for IP Multicast problems, ALL switches on ALL systems, regardless of software revision, should be configured to disable snooping.

2 System Planning and Analysis

2.3 PCN Network Configuration and Components

3. Domain Planning Guidelines

3.1 Basic concepts of a Windows domain



REFERENCE - EXTERNAL

For detailed description about the Windows domain and the Security concepts, refer to Microsoft® documentation.

Overview of Windows domain

The Windows domain is a group of computers running versions of Microsoft Windows operating system. The operating system shares a central database (also known as the Active Directory). The central database contains the user accounts and security information for the resources in the domain. Each computer user of a domain has unique account or user name. The user accounts have rights/permissions to access the resources in the domain.



REFERENCE - INTERNAL

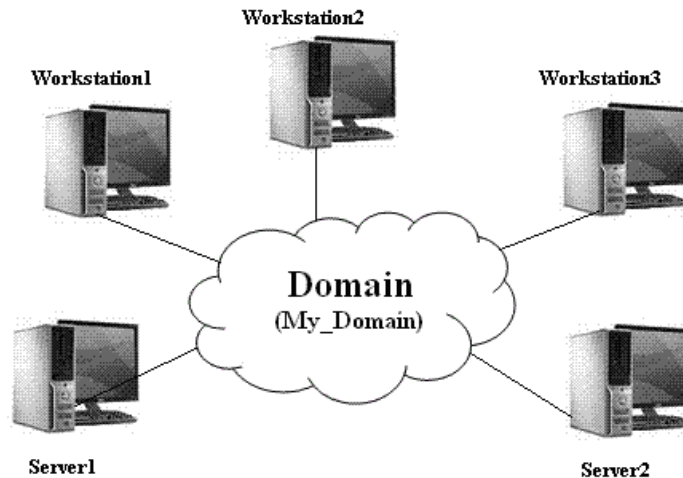
For details about the Active Directory, see “Overview of Active Directory”.

3 Domain Planning Guidelines

3.1 Basic concepts of a Windows domain

The following figure shows a typical Windows domain:

Figure 1 Windows domain



Need for a Windows domain

The following are the benefits of connecting multiple computers to a domain:

- **Centralized Administration:** The database containing the details of the resources in a domain is centralized. This ensures that the entire domain is managed accessing single database.
- **Single Logon process:** Through single logon
 - grant access to the network resources
 - assign the permission
- **Scalability:** You can create large networks.



REFERENCE - INTERNAL

For details about creating a Windows domain, refer to the *TPS System Implementation Guide*.



TIP

The alternative to a domain is a Windows Workgroup.

A Windows Workgroup is a group of computers running versions of Microsoft Windows operating system.

The computers in a Workgroup are standalone. The Workgroup environment requires that policies be implemented on each individual computer. The domain environment allows policies to be implemented across entire groups of computers and users with one action.

The Workgroups model is considered difficult to manage, if there are more than five computers, and lacks benefits of a domain.

Benefits of having TPS nodes in a Windows domain

The following are the benefits of connecting TPS nodes to a Windows domain:

- Administration: The database containing the details of the resources in a domain is centralized.
 - Centralized database simplifies management of the TPS user accounts such as engineers, supervisors, operators, administrator, and so on. For the TPS nodes in a domain, maintain a copy of the TPS user accounts in the Domain Controller. In the absence of a domain, maintain the user account in every node. For example, if there are twenty users using a TPS system of 10 nodes, then each node must have a copy of twenty user accounts.
- Security: Through single logon, grant access to the network resources and assign the permission.
 - Assign membership of users to the appropriate user groups. Assign GPOs to these groups. This helps in managing the user's access privileges. To change the access privileges of a user group, change the corresponding security settings of the Group Policy in the Domain Controller. For example, the Operators and Supervisors cannot access the internet. However, if the user uses WebGUS, they have to access internet through a specific proxy. To grant access, configure the proxy for web access in the Group Policy for Operators and Supervisors. Therefore, the users of Operators and Supervisors group on every TPS node receive the new policy on subsequent logins.
- Scalability: You can create large networks.

3 Domain Planning Guidelines

3.1 Basic concepts of a Windows domain

GUS

The following nodes are not required to be a member of a network:

- USR (US replacement) nodes that runs Native Window only
- Standalone GUS nodes that runs GUS and Native Window

The networked GUS that runs GUS and Native Window, are not required to be a member of a Windows domain.

The TPS GUS node that runs GUS and Native Window must be a member of a Windows 2003 domain.

APP

The AMR (AM replacement) nodes that run basic AM functionality are not required to be a member of a network.

The TPS APP node that runs AM and APP functionality, must be a member of a Windows 2003 domain.

Overview of Trees and Forests

The number of domains required for a company can be more than one. If there is one or more domain, group the multiple domains to form structures called trees and forests.

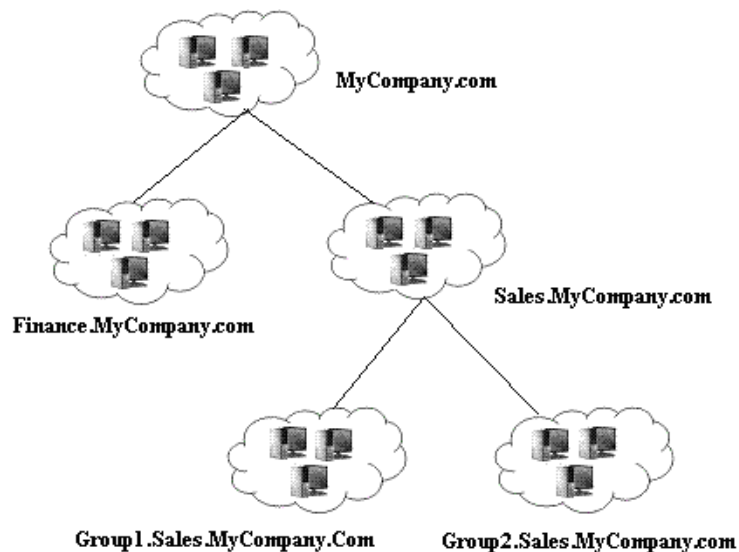
Trees

A tree is a collection of domains. The tree structure starts with a single root domain and branches out into child domains. The first Active Directory domain created becomes the root of the first tree structure. The other domains created later become the child domains. The child domain can become the root domain of a new tree.

The name of the tree is always the DNS name of the domain at the root of the tree. The child domains of the root domain are always contiguous with the root domain in the namespace.

The following figure shows the contiguous namespace of a tree structure:

Figure 2 Contiguous namespace of a tree



3 Domain Planning Guidelines

3.1 Basic concepts of a Windows domain

The following are the reasons for creating multiple domains in a domain:

- To manage different organizations or to provide unit identities
- To enforce different security boundaries and password policies
- To control the Active Directory replication
- To handle large number of managed objects
- To de-centralize administration

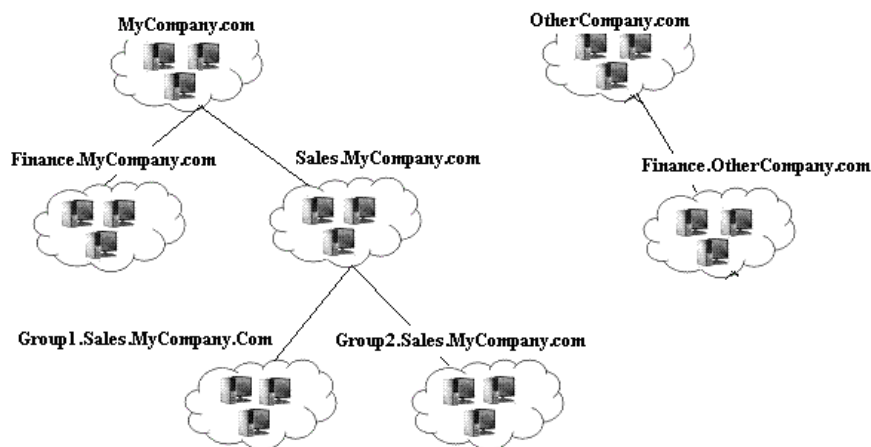
Forests

The multiple domains required for a company can be non-contiguous. A forest structure permits creating multiple child domains that are non-contiguous with the root domain in the namespace. The characteristics of a child domain in a forest structure are:

- non-contiguous with the root domain
- operates independently
- belongs to the same network

The following figure shows the non-contiguous namespace of a forest structure:

Figure 3 Non-contiguous namespace of a forest



3.2 Domain Controller concepts

Overview of Domain Controller

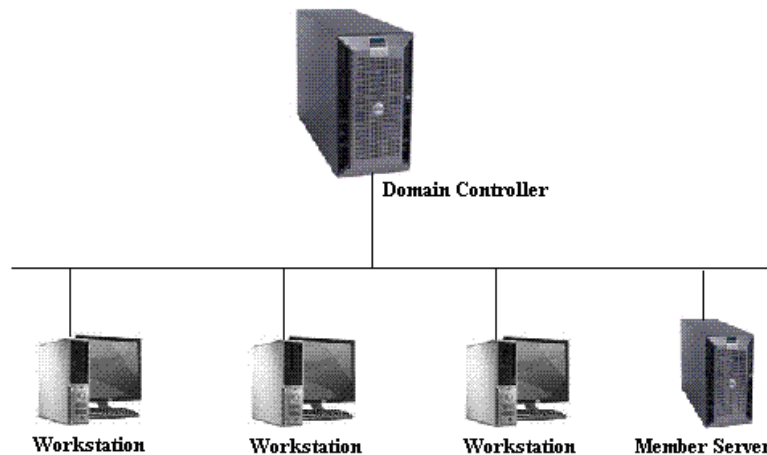
The Domain controller is a server machine that:

- Runs Windows 2000 Server or a Windows Server 2003 operating system
- Stores the read-write copy of the Active Directory and the security information of a particular domain
- Manages communication between the users and the domains including:
 - User logon processes
 - Authentications
 - Directory searches

There is one Domain controller in every Windows domain.

The following figure shows the domain controller in a Windows domain:

Figure 4 Domain Controller



3 Domain Planning Guidelines

3.2 Domain Controller concepts



REFERENCE - INTERNAL

For details about implementing a Domain Controller, refer to the *TPS System Implementation Guide*.

Overview of Member Server

The Member Server is a computer that:

- Runs Windows 2000 Server or a Windows Server 2003 operating system
- Connected to a domain
- Not a Domain Controller

A Member Server functions as the following types of servers:

- File servers
- Application servers
- Database servers
- Web servers
- Certificate servers
- Firewalls
- Remote access servers

The Member Server does not perform the following functions:

- Domain account logon
- Active Directory replication
- Domain security policy information storage

Overview of Peer Domain Controller

A domain can have more than one domain controller. You can add additional domain controllers (Peer Domain Controller) over the network or from a backup media. Multiple domain controllers in a domain provide the following benefits:

- Improves the availability and reliability of network services
- Provides the fault tolerance
- Balances the load of existing domain controllers
- Provides additional infrastructure support to sites
- Improves the performance by allowing the user to connect to a domain controller when logging on to the network
- Continuous functioning of the domain (when one of the domain controllers fail or must be disconnected, the other domain controller becomes the active domain controller)



REFERENCE - INTERNAL

For details about implementing a Peer Domain Controller, refer to the *TPS System Implementation Guide*.

3 Domain Planning Guidelines

3.3 Domain Controller Capabilities in the TPS System

3.3 Domain Controller Capabilities in the TPS System

Windows domain controller use in a TPS system

The TPS system infrastructure leverages Windows domain controller concepts for the underlying system. In general, a Windows domain simplifies the administration and security of member nodes and this is why it is embraced as part of the TPS system. To establish a Windows domain, at least one domain controller must exist on the network.



ATTENTION

Windows 2000/2003 releases of Experion Server - TPS (ESVT), Experion Station - TPS (ES-T), GUS, and APP require a Windows domain controller.

Additional domain controllers increase reliability

After the initial Windows 2003 domain controller is installed and operational, consider installing additional domain controllers to increase reliability. In the event of a domain controller failure, other domain controllers within the same domain can provide the authentication duties. Unless otherwise specified, authentication takes place on the domain controller with the fastest access to the client machine. As mentioned earlier, each domain controller is an equal partner or peer with the others. DC security information is shared through Windows 2003 multi-master replication. Therefore, administration tasks such as adding a domain user can occur as long as at least one domain controller is available.



ATTENTION

Honeywell recommends that a basic Plant Control Network include at least two domain controllers to maintain a reliable system. If possible, the domain controllers should also be physically secure and separated for additional fault tolerance. Sites with multiple control rooms should implement at least one domain controller in each control room. Review the Domain Controllers section for additional recommendations.

Domain controller capabilities

The following table details some of the capabilities of a domain controller:

Table 7 Domain Controller Capabilities

Item	Description
Establishes a security boundary	All users outside the domain are considered untrusted and denied access unless access is explicitly granted individually or through legacy or transitive trust relationships.
Single Location for User and Group accounts	Users define accounts and account attributes only once rather than on each machine. Again, this is similar to the current operation in established process control systems.
Define and manage security of the TPS nodes	The TPS system security model is configured by the user and distributed to each node in the network. This security model is used in the same way keylock access is used in the TPN System.
Distribute common files	GUS display files, SafeView, configuration files, and so forth, can be stored on a domain controller. These files can be distributed to the network nodes through Windows 2003 "offline folders" capability. Note: Configuring Windows 2003 Offline Files is not required for the proper operation of a TPS system.
Perform central software backups	A domain controller also stores the TPS domain information. Note: The TPS domain is a logical grouping of nodes for use in the TPS system. TPS system software uses the Organizational Unit concept to represent TPS domains.

3 Domain Planning Guidelines

3.4 Active Directory Guidelines

Minimum requirements for domain controller

Following is a list of the minimum hardware and software configuration requirements for a basic Domain Controller:

Item	Description
Computer and processor	PC with a 133-MHz processor required; 550-MHz or faster processor recommended. (Windows Server 2003 Standard Edition supports up to four processors on one server)
Memory	128 MB of RAM required; 256 MB or more recommended; 4 GB maximum
Hard disk	1.25 to 2 GB available hard-disk space
Operating System	Windows 2003 Server



ATTENTION

Honeywell recommends that for all but the simplest environment (GUS as a US replacement or APP as an AM replacement), a domain controller is a definite system requirement. The domain controller must meet minimum system requirements and be physically secure.

3.4 Active Directory Guidelines

Overview of Active Directory

The computers in a Windows domain share a database called the Active Directory. The database is centralized, organized, and accessible to the resources of the domain. The Domain Controller stores the resource information and the security settings of an organization in the Active Directory.

The Active Directory is a directory service that performs the following functions:

- As a directory stores information about users and resources
- As a service or services provides access to manipulate the resources

The Active Directory manages all elements of the network; including computers, groups, users, domains, security policies, and any type of user-defined objects.

Active Directory purpose

You can think of an Active Directory as a distributed database that can have enterprise scope if configured as such. An Active Directory provides distributed security, user, group, and computer management dynamic name services. The following sections describe the basic components and capabilities of an Active Directory.

Forest and tree domain concepts

The Active Directory includes domain concepts including forests and trees. Trees are the hierarchical representation of domains that share a contiguous DNS namespace. Domains that are members of a tree are established with a parent-child relationship. At each level of a tree (except the root), a domain maintains a two-way trust relationship with its parent domain. The transitive nature of the trust relationship allows each domain to maintain a two-way trust with every other domain in the forest. This trust relationship is established automatically upon creation of a child domain.

Since Windows 2003 and DNS are closely integrated, typically you can discern the domain architecture through its DNS name. For example, node1.development.mydomain.local indicates that node one is a member of the development domain, which is a child of mydomain. Mydomain is the root of the tree.

A forest is a collection of domain trees that share a common schema (see section 3.5, Dynamic DNS Role in Process Control Domain). Domain trees within a forest may not share a contiguous DNS namespace; however, they do continue to maintain a full two-way transitive trust relationship with each other.

Process control domain considerations

Careful planning and coordination with IT should be in place prior to adding a process control domain to an existing enterprise IT forest. The process control administrator should also consider the following:

- Schema and configuration changes have forest-wide impact.
- Child domains maintain two-way trust relationships with all the other domains in the forest.



ATTENTION

The following table lists Honeywell's recommendations for configuring Active Directory based domain architectures in the process control environment.

3 Domain Planning Guidelines

3.4 Active Directory Guidelines

IT Enterprise Domain Configuration	Recommended Windows 2000/2003 Process Control Domain Architecture	Description
None	Single Forest for each Process Control Domain	This scenario typically corresponds to a process control implementation isolated from the IT enterprise.
Windows 2000/2003	Single Forest for each Process Control Domain	If the IT organization has a Windows 2000/2003 infrastructure, you may still want to implement process control domains in separate forests to address issues such as security and schema modifications.
Windows 2000/2003	Child Domain	If the IT enterprise infrastructure supports Windows 2000/2003 Active Directory and the site is looking for tight interoperability between IT and Process Control domains, configure a child domain. This option is typically selected for customers whose IT organizations maintain process control domain architectures. Note: Carefully consider the previously described implications of creating a child domain.

Schema defines Active Directory objects

The schema represents the definition of all Active Directory objects. An example of the schema is the definition of a user object that can include attributes such as First Name, Last Name, Phone Number, and Email Address. The object, on the other hand, would be the actual data that populates each of the properties: John Smith, 555-1212, and john.smith@mydomain.com. The scope of a schema is an entire Windows 2003 forest. A change to the schema (for example, extending the definition of a User to include the home phone number) will result in updates to all domain controllers in all of the domains in the forest.



ATTENTION

Honeywell recommends that prior to joining an existing forest, become familiar with your IT enterprise schema change control mechanisms. In addition, carefully evaluate and understand the implications of replicating an Active Directory.

Place process control domains into unique sites

Adding, deleting, or modifying objects within the Active Directory will initiate replication traffic. Object replication is bounded by a Windows 2003 domain. However, a subset of each object is replicated to other domains through the Global Catalog. If process control domains are included within a larger enterprise-level forest, consider placing process control domains in a separate Windows 2003 site and decreasing replication traffic. Windows 2003 sites define well-connected networks and are able to govern replication traffic. See Microsoft or other appropriate documentation for further information.



ATTENTION

Honeywell recommends that for process control domains that are members of a larger IT enterprise forest, limit the effect of replication traffic on process control domains by placing process control domains in unique Windows 2003 Sites and limit the frequency of Active Directory replications.

3.5 Dynamic DNS Role in Process Control Domain

Overview of Domain Name Service

The Domain Name Service (DNS) is a service that translates the domain names into IP addresses. An IP address is a series of four groups of numbers separated by periods. A domain name is a series of alphanumeric strings separated by periods. The IP addresses identify the computers in a network. If you refer a computer using the domain name (for example, mycompany.com), the DNS service translates the name into the corresponding IP address (for example, 100.10.0.123).

The DNS is installed in the Domain Controller (the first and only domain controller installed in the domain). This Domain Controller helps in resolving the domain/computer names between the computers in the domain.

The users of a domain do not communicate directly with a DNS resolver. However, the domain/computer name is resolved in the client applications like web browsers, mail clients, and other Internet applications. When there is a request that needs a DNS resolution, the following actions are performed:

- The applications send a resolution request to the local DNS resolver in the operating system.
- The local DNS resolver sends the request to the designated DNS server or servers.
- The DNS server provides the correct IP address.



ATTENTION

In a domain tree, if one DNS server is not able to resolve the domain/computer name, it communicates with another DNS server to obtain the correct IP address.

DNS integrated with Active Directory

The integration of DNS with the Active Directory is a central feature of the Windows 2003 Server operating system. DNS is so closely tied to Windows 2003 that a DNS server will be automatically installed and configured on the first Windows 2003 domain controller if a Dynamic DNS-capable server is not identified.

Significance of DNS naming conventions

Each node and domain is represented by a DNS-compliant name. A fully qualified DNS name in a Windows 2003 environment reveals the hierarchical domain structure. For example, “node1.development.mydomain.com” indicates “node1” is a member of the “development” domain that is a child of “mydomain.” “Mydomain” is the root of this domain hierarchy.



ATTENTION

DNS permits only a subset of the valid legacy NetBIOS name characters; therefore, Windows 2003 migrations may require changes to computer host names to conform to the DNS conventions.

The user should carefully consider domain-naming conventions, because domain names cannot be modified after they are created. One recommended naming convention for domains is that members of the overall IT Windows 2003 enterprise infrastructure are to append “.local” to the end of the domain name.

For example, if process control domain “PC” is created in an enterprise using a Windows 2000-based infrastructure, the root domain name could be represented as “PC.local.” A second process control domain, “PC2,” could use the recommended convention and use the name “PC2.local.” In each case, a Dynamic DNS server is established that resolves names in each of the local domains only. These local dynamic DNS servers should have forward references to the existing enterprise DNS servers. The forward references allow local process control domain node members to resolve machine names in the IT enterprise.

Honeywell recommended DNS naming conventions



ATTENTION

Honeywell recommends determining whether Process Control or IT will own dynamic DNS servers. If IT establishes a Windows 2003 infrastructure, then Process Control can leverage the IT dynamic DNS servers.

Process control domains that are implemented as their own forest should use the “local” naming convention.

3.6 TPS Domain Planning

Role of TPS domains

TPS domains provide a finer granularity namespace than a Windows 2003 domain. The TPS system implementation of TPS domains on Windows 2003 uses the Organizational Unit capabilities in Active Directory. Earlier TPS releases supported the TPS domain residing within a single Windows domain, and were implemented using replicated registry keys.

Overview of Organizational Units

An OUs is an Active Directory container. You can place the domain objects like users, groups, computers, and so on in the OUs. An OU cannot contain objects from other domains.

The domain for any organization can enlarge and becomes difficult to manage. The organization can breakdown a very large domain into smaller units for easy management. The Active Directory uses the OUs to manage large domains.

You can arrange the OUs hierarchically in a tree-like structure. The tree structure must match with the organization structure for security boundaries or ease of administration.

The organization can divide its large domain into OUs based on their department. For example, within business.com, an OU can be created each for Sales, Support, Marketing, Development, and Q/A. An OU within it can contain another OU. The organization can extend the hierarchy of OUs, as required by the organization's hierarchy within a domain. The OUs created in a domain helps to reduce the number of domains required for a network.

The OUs can delegate the administrative control over them to a subset of users in the Active Directory. For instance, the domain administrator needs to designate one person in each department as the official Password Change Administrator. This reduces the administrative load. The domain administrator can delegate the authority to modify users' passwords to each user over only their respective OU. The administrator authenticates the users, however restricts them to certain areas of the Active Directory infrastructure.



REFERENCE - INTERNAL

For details about creating Organizational Units, refer to the *TPS System Implementation Guide*.

TPS domains as Organizational Units (OU)

TPS domains are created as Windows 2003 Organizational Units (OUs). You use the Windows 2003 MMC Users and Computer snap-in to administer OUs. Administration includes the creation and deletion of OUs and the addition of users, groups, and computers as members of OUs.

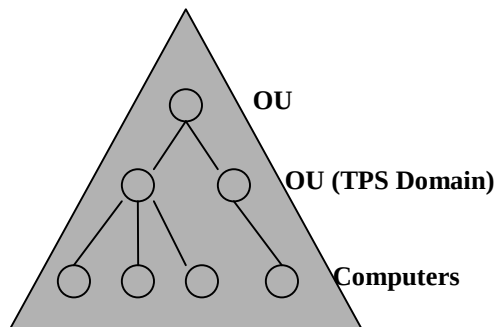
A Windows 2003 Organizational Unit is a container object within the Active Directory. An OU may contain Active Directory objects such as computers, users, and groups or additional nested OUs. The nesting capability provides the framework for implementation of a hierarchical namespace. OUs are also tightly integrated into Windows 2003 security infrastructure. Security policies can be applied on the basis of OU membership. In addition, administrative capabilities can be delegated to the administrator for a specific OU.



REFERENCE - INTERNAL

For more information on Organizational Units, see the *System Management Configuration Guide*, section "What is a TPS Domain or Console?"

The following diagram illustrates a sample domain architecture consisting of nested OUs:



TPS domain configuration requirements

TPS domain configuration is performed using the Active Directory Users and Computers MMC snap-in in conjunction with the TPS domain configuration extension snap-in. The extension allows you to create an Organizational Unit in the Active Directory with an attribute of a TPS domain.

3 Domain Planning Guidelines

3.6 TPS Domain Planning

Determining TPS domain configurations

After you decide on the Windows 2003 Domain configuration, you need to plan the number of TPS domains and the number of nodes within each TPS domain. A TPS domain typically represents a single TPN; but you could also partition a TPS Domain into the scope of control of one or two operators. Honeywell Communication Interface (HCI) communications are not bounded by TPS domains as in previous releases.

Nodes that communicate with each other through HCI are usually in the same TPS domain, but are not restricted to the same TPS domain. Non-TPS client configuration allows nodes with no TPS domain affiliation to communicate with members of a TPS domain.

Node component configuration architecture

The current release of the TPS system's component configuration architecture does not use the domain controller as a central repository. Component information is stored only on the node that hosts the component. Replication of HCI component information is not necessary, because remote components are known to the local node through the HCI Name Service. Additionally, component-specific files, such as HCI security proxy files, reside only on the node that hosts the component.

In the event of a catastrophic node failure, such as a hard disk failure, HCI component configuration will be lost. In the absence of a current backup, you will be required to manually reconfigure the local components after reloading the applicable software.



ATTENTION

Component configuration is automatically saved when a component is initially configured; and HCI components are automatically checkpointed prior to a component shutdown. Honeywell recommends that you checkpoint after making database changes.

A checkpoint does not save the component configuration in the node's registry. To avoid losing component configuration in the event of a catastrophic node failure, use conventional third party backup utilities.

3.7 Sample TPS System Topologies

Native mode recommended



ATTENTION

Honeywell recommends that you configure your systems in a Windows 2003 native mode. Several scenarios follow that describe mixed mode operation and are intended to assist you with planning for Windows 2003 native mode architecture.

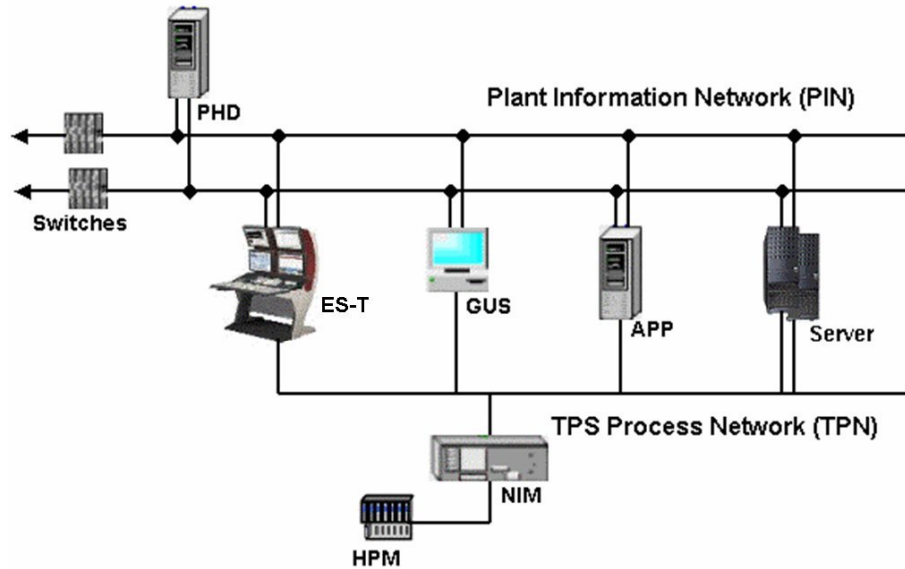
Sample TPS system topology - FTE supports a Fault Tolerant Community

Fault Tolerant Ethernet (FTE) is Honeywell's intermediate network driver that provides fault detection and recovery capability using commercial-off-the-shelf Ethernet products for a Plant Control Network (PCN).

You can organize a PCN into one or more Fault Tolerant Communities. Within a Fault Tolerant Community, errors are detected rapidly and recovery is enabled. Fault Tolerant Communities do not have to correspond to Windows domains.

The following diagram illustrates a basic FTE-enabled LAN configuration:

Figure 5 Fault Tolerant Community

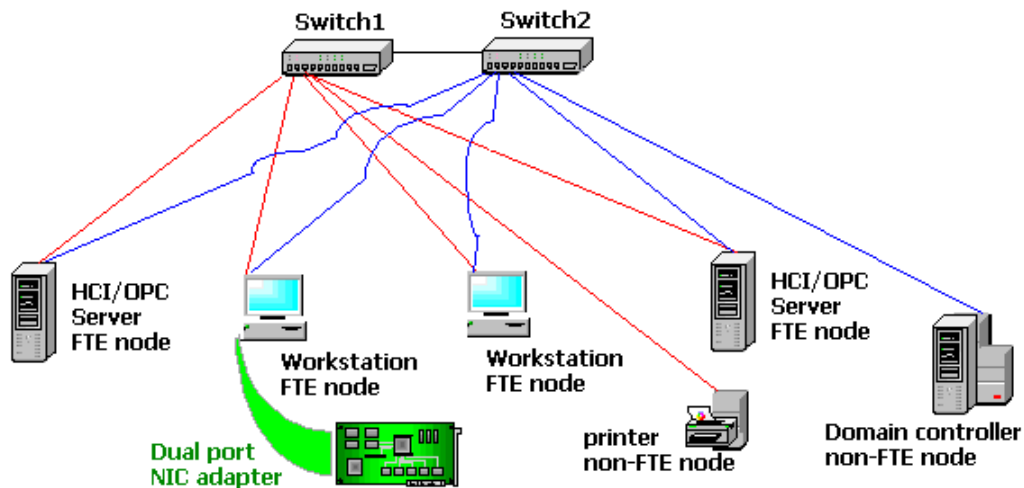


An FTE-enabled LAN contains redundant network components, such as switches, cables, and NIC adapters. Nodes with redundant NIC adapters and the Honeywell FTE software are able to participate in the Honeywell Fault Tolerant Ethernet.

FTE-enabled nodes are resilient to single Ethernet failures, such as switch or cable faults. Client applications in FTE nodes are entirely abstracted from the underlying Honeywell Fault Tolerant Ethernet. Two types of hosts, FTE nodes and non-FTE nodes, may reside on a FTE-enabled LAN.

FTE nodes connect to the LAN using dual port NIC adapters. Each adapter port has a unique IP address. When operational, each FTE node sends diagnostic messages to determine the health of the LAN in addition to identifying the best available network path for peer-to-peer communication. Non-FTE nodes are singly connected to the LAN configuration, but do not run FTE software and are not able to take advantage of Honeywell Fault Tolerant Ethernet.

Figure 6 FTE in Network Segment



ATTENTION

Windows 2003 domain controllers (including the WINS server) must not be FTE nodes. The root domain controller should be singly connected to tree A and the peer domain controller to tree B of an FTE network; or the domain control function should be supplied from the other side of a firewall.



REFERENCE - INTERNAL

For additional details about using FTE in a TPS system topology, refer to the following documents:

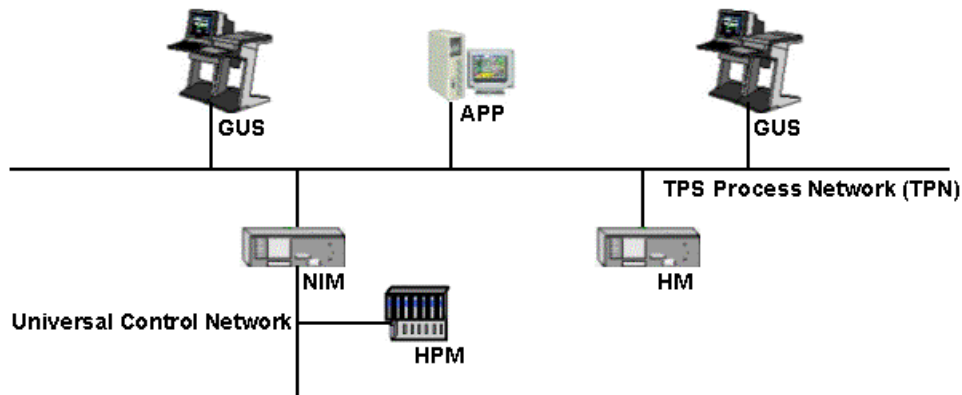
- [FTE Overview and Implementation Guide](#)
- [FTE Planning, Installation and Service Guide](#)

Sample TPS topology - US/AM replacement using GUS and APP

In the following diagram shows a US/AM replacement topology where a GUS node is used instead of a TPN Universal Station, and an APP is used instead of a TPN Application Module. Each node is self-contained and separately administered. At least one removable media drive is required for each console. The AM replacement is operationally similar to the current AM as all files are obtained from the History Module (HM).

Each standalone GUS must have a removable media device for transferring GUS displays to that node. This US/AM replacement topology does not require a PCN networking infrastructure.

Figure 7 US/AM Replacement Using GUS and APP



Sample TPS topology - US/AM replacement using Experion nodes

The US/AM replacement topology shown in the following diagram uses Experion Station TPS (ES-T) instead of Universal Station (or GUS), and uses an APP node instead of an Application Module (AM).

The APP node is operationally similar to the AM – it obtains all files from the History Module (HM).

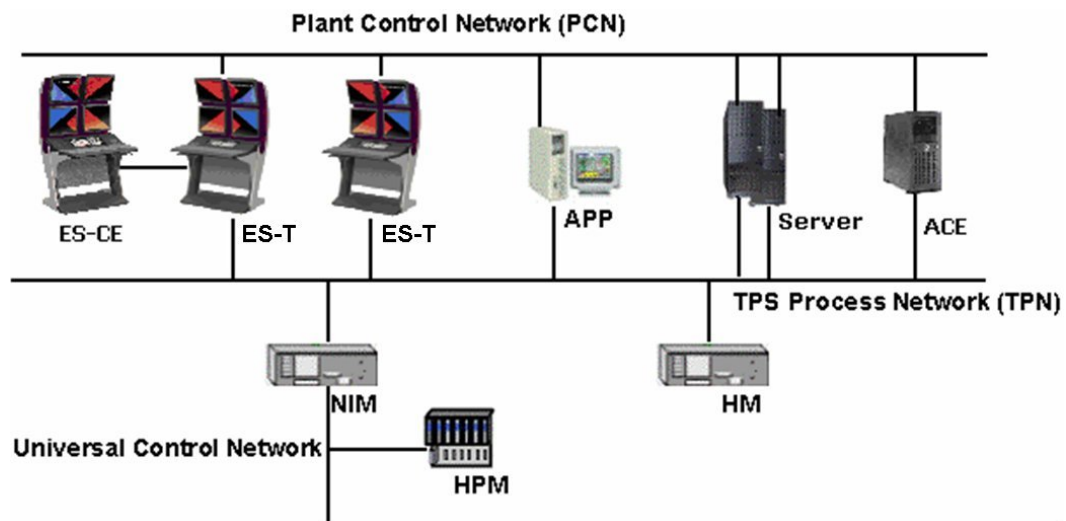
A TPN-connected Server (ESVT), a single or redundant pair, provides repository services for Experion components, process and event management, and OPC Server functions.

A TPN-connected Application Control Environment (ACE -T) provides all control functions through its added LCNP card.

An Experion Station - Console Extension (ES-CE) allows a station to participate in a Console without being connected to the TPN.

A US/AM replacement topology that uses TPN-connected Experion nodes (ES-T, ESVT, or ACE-T) requires a PCN network infrastructure.

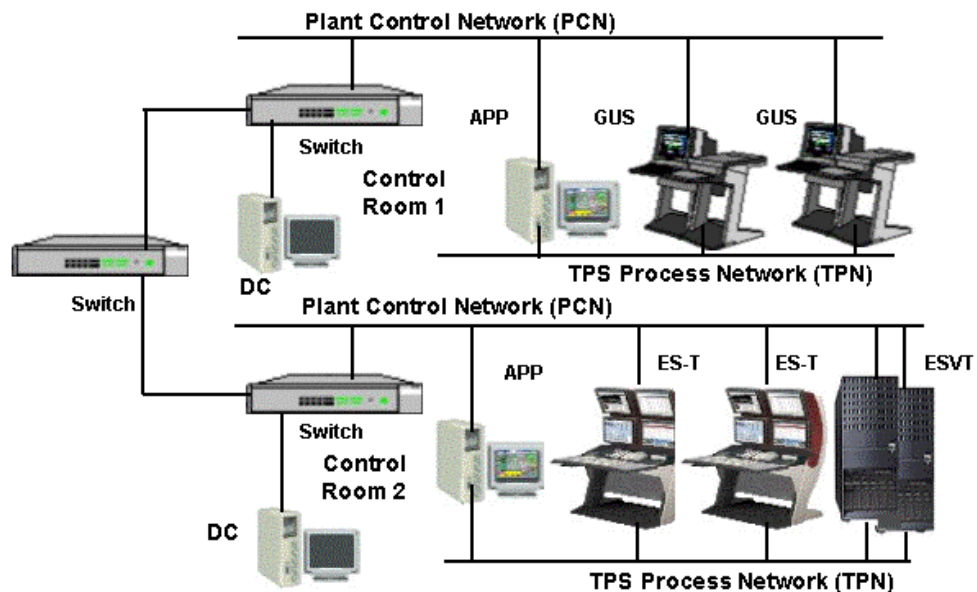
Figure 8 US/AM Replacement Using Experion Nodes



Sample TPS topology – plant site with multiple control rooms

The following diagram shows a plant site with multiple control rooms. Peer domain controllers are added for each control room. The additional domain controllers provide fault tolerance and faster login authentication. Typically, you would implement this configuration as two separate TPS domains (one for each TPN segment).

Figure 9 Plant Site with Multiple Control Rooms

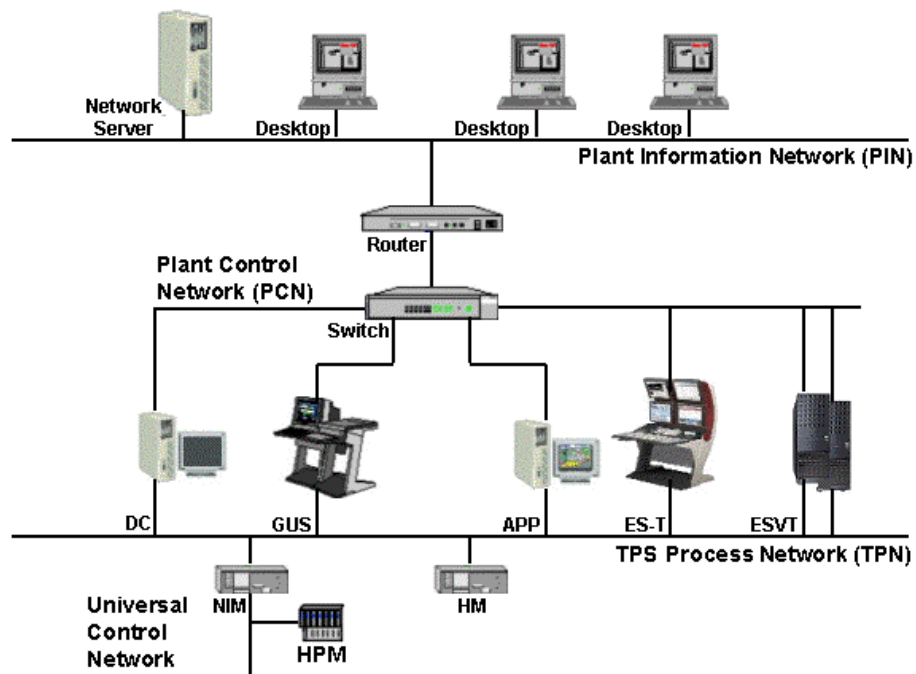


Sample TPS topology– Router to provide PCN connection to PIN

The following diagram shows the addition of a router to extend basic PCN configuration to provide PIN connectivity. The router provides a secure and single point of entry onto the PCN. Additional security measures may include the use of firewall software.

The business domain and the process domain would typically maintain a one-way legacy Windows NT 4.0 trust relationship if the business environment were Windows NT 4.0-based. If, on the other hand, the IT environment is Windows 2003-based, the process control environment could be created as a Windows 2003 child domain. In both cases, clients on the business domain could use capabilities such as OPC or Remote Native Window to view process control data.

Figure 10 PIN Connectivity



ATTENTION

If the system includes Experion nodes, the business/IT network must be Windows 2003.

Sample TPS system topology – PIN connectivity and Uniformance

The following diagram shows the key capabilities of the PIN connectivity and Uniformance topology. This example builds upon the capabilities shown in the previous diagram. The following table describes the capabilities illustrated in Figure 11:

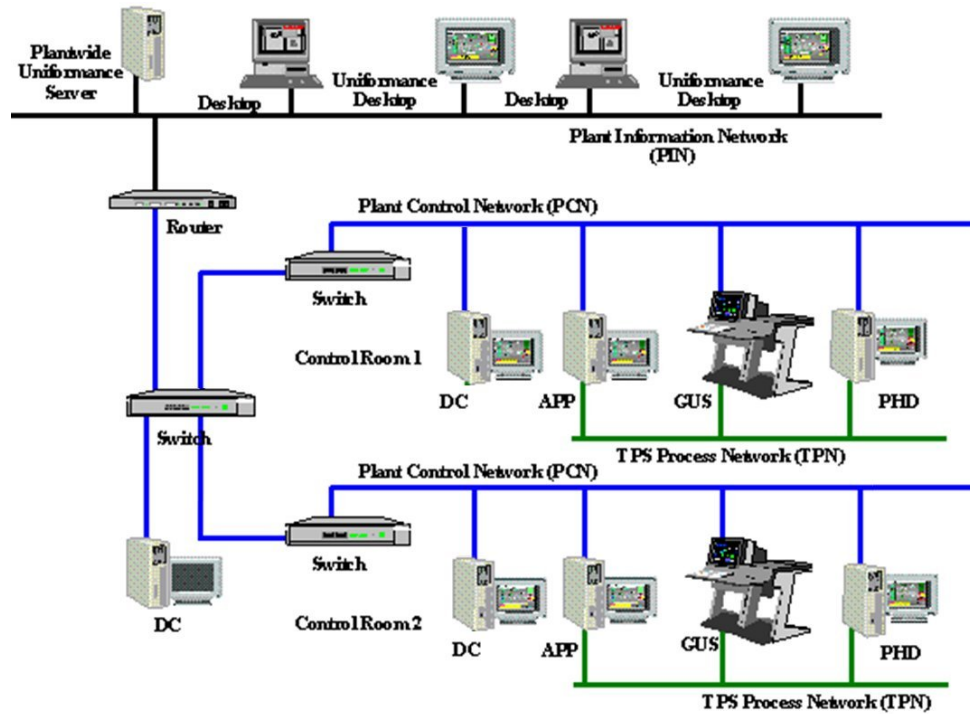
Table 8 PIN Connectivity and Uniformance Components

Item	Description
Backbone Switch	Backbone switches provide control room-to-control room connectivity. Backbone switches may also require high bandwidth options such as Gigabit Ethernet. For additional details, see section 5, <i>Industrial Strength Networking Concepts</i> .
PHD Servers	A PHD Server for each TPN is typically implemented to capture history data.
PHD Shadow Server	The shadow server consolidates plant-wide data from the individual PHD servers. The diagram shows the shadow server residing on the PIN to expose data to enterprise applications and business clients.

This configuration provides two ways for PIN clients and applications to access data:

- Communicate with the process control components in the Process Control Network through capabilities such as Remote Native Window or OPC.
- Communicate with the plant-wide Uniformance server on the PIN. This method can reduce the amount of PIN traffic on the PCN.

Figure 11 PIN Connectivity and Uniformance



3.8 Users, Computers, and Groups

User Account

A user requires an Active Directory user account to log on to a computer or to a domain. This account provides an identity for the user. The operating system uses this identity for the following purposes:

- To authenticate the user
- To grant access privileges to specific domain resources

To enable user authentication and authorization features, perform the following:

- Create an individual user account for each user on the network.
- Add each user account to the groups.
- Assign appropriate rights and permissions to each group.



TIP

You can use the user accounts as a service account for some applications. You can configure a service to log on as a user account. The user account grants access to the service, to access the specific network resources.

Computer Account

A Windows computer must require a unique computer account to access the resources in a domain. The computer account provides the following:

- Authenticates the computer to access the network
- Audits the computer's access to the network and the domain resources

Groups

A Group is an Active Directory objects. The Group can contain users, contacts, computers, and other groups.

The following are the two different types of Groups:

- Distribution Groups
- Security Groups

Distribution Groups

The Distribution Groups creates the e-mail distribution lists. The e-mail applications (like Microsoft Exchange) use the distribution group to send e-mail to the members of the group. As with a security group, add a contact to a distribution group so that the contact receives e-mail sent to the group.



ATTENTION

Honeywell does not recommend the usage of e-mail on Process Control Domain used by the Experion and TPS.

Security Groups

The Security Groups are the important component of the relationship between the users and the security. The Security Groups perform the following functions:

- Manages the user and the computer access to the shared resources
- Filters Group Policy settings

You can collect the objects such as users, computers, and other groups into a Security Group. The Security Group has access privileges to access the specific resources. You can assign access privileges to the Security Group therefore, it is not necessary to repeat for individual user. When you add a new user to the group, the user receives all access privileges assigned to the security group.



REFERENCE - INTERNAL

For details about creating User Account/Computer Account/Group, refer to the *TPS System Implementation Guide*.

3.9 Group Policy

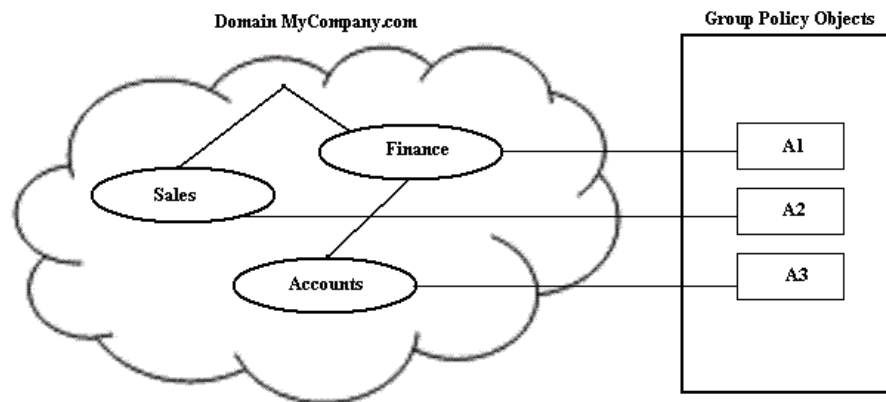
Overview of Group Policy

Group Policy is an infrastructure used for delivering and applying one or more configurations/policy settings to the users and the computers within an Active Directory environment. The Group Policy Objects (GPOs) have the Group Policy settings. You can link the GPOs in a domain to Domains or OUs.

An organization can have different types of users. You must deliver and maintain a customized desktop configuration for different types of users, including mobile users, information workers, or others assigned strictly to defined tasks. For example, one class of users does not have access to Internet Explorer, but the other class of users can access Internet Explorer. Group Policy helps in applying a customized configuration to a group of users.

The following figure shows the customized group policies assigned to the OUs within a domain:

Figure 12 Group Policy objects



You can infer the following from the preceding figure:

- The Group Policy A1 is applied to the Finance OU.
- The Group Policy A2 is applied to the Sales OU.
- The Group Policy A1 and A3 are applied to the Accounts OU.
- The members of the OUs receive the Group Policy from the respective OUs.

You can link the GPOs to the sites, domains, and OUs to implement the Group Policy settings. You can apply the settings to the organization or to a portion of the organization.

The GPO links affect the users and the computers in the following ways:

- A GPO linked to a domain applies to all users and computers in the domain. The users and computers of the child OUs also inherit the GPO.
 - The policy is not applicable across the domains.
- A GPO linked to an OU applies to all users and computers in the OU. The users and computers of the child OUs also inherit the GPO.

Overview of Group Policy Settings

The Group Policy includes the following types of policy settings:

- User Configuration Settings
- Computer Configuration Settings

User Configuration Settings

The User Configuration Settings contain the policy settings that affect users, regardless of which computer they log on to.

The following are the user-related policies specified in the User Configuration settings:

- Operating system related settings
- Desktop settings
- Application settings
- Security settings
- Assigned and published applications options
- User logon and logoff scripts
- Folder redirection options

Applies the user-related policy settings:

- when the users log on to the computer
- during the periodic refresh of the Group Policy

3 Domain Planning Guidelines

3.9 Group Policy

Computer Configuration Settings

The Computer Configuration Settings contain the policy settings that affect computers, regardless of who logs on to the computers.

The following are the computer-related policies specified in the Computer Configuration settings:

- Operating system behavior
- Desktop behavior
- Application settings
- Security settings
- Assigned applications options
- Computer startup and shutdown scripts

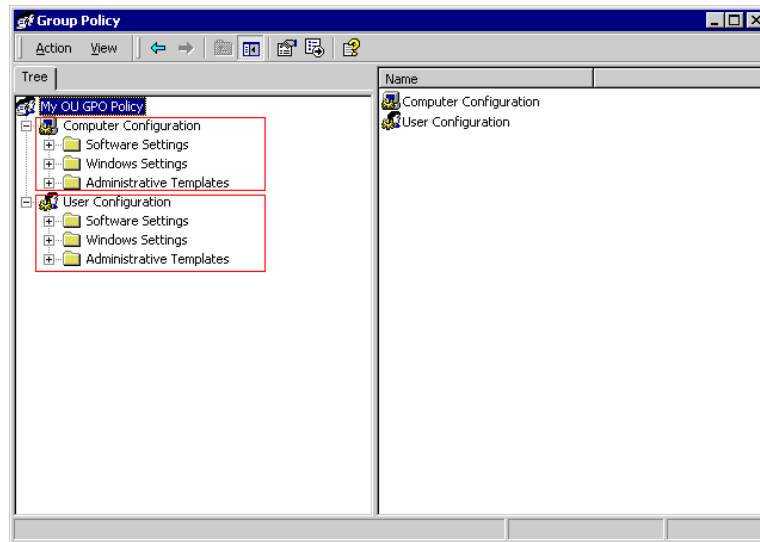
Applies the computer-related policy settings:

- when the machine is restarted
- during a periodic refresh of the Group Policy

You can open the GPO in the Group Policy Editor to view and edit the Group Policy Settings.

The following figure shows the GPO opened in the Group Policy Editor:

Figure 13 Group Policy editor



The settings under 'Computer Configuration' are applied to all computers. You can edit this group of settings and apply to a group of computers.



ATTENTION

The GPO with settings limited to computer configuration does not have any effect when it is applied to a user.

The settings under 'User Configuration' are applied to all users. You can edit this group of settings and apply to a group of users.



ATTENTION

The GPO with settings limited to user configuration does not have any effect when it is applied to a computer.



REFERENCE - INTERNAL

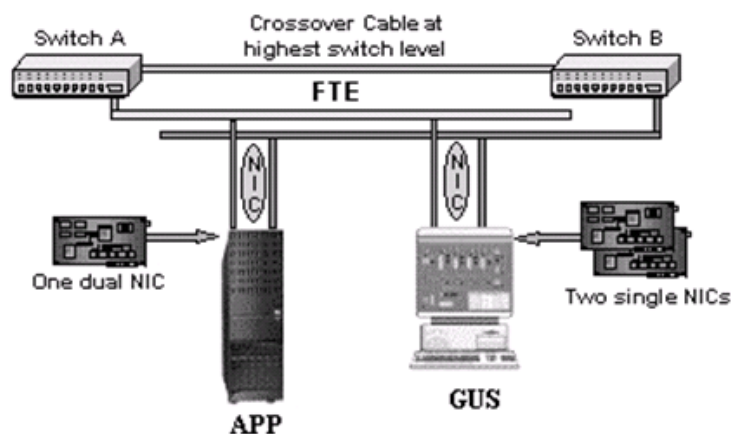
For details about creating Group Policy, refer to the *TPS System Implementation Guide*.

3.10 FTE

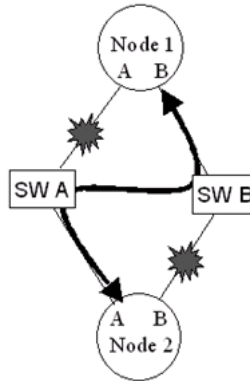
Overview of FTE

Honeywell's Fault Tolerant Ethernet (FTE) is a redundant network topology. Honeywell's FTE driver and commercially available components accomplish its redundancy. When the primary route fails, the network communication occurs over an alternate route using the driver and the FTE-enabled components. Each FTE node connects twice to a single LAN through the dual Network Interface Card (NIC) as shown in the following figure:

Figure 14 FTE



When the failure occurs, the FTE continues to communicate as shown in the following figure:



There are broken channels on FTE Node 1 (Channel A) and FTE Node 2 (Channel B). However, the nodes continue to communicate from FTE Node 1's Channel B to FTE Node 2's Channel A.



REFERENCE - INTERNAL

For details about FTE, refer to the following documents:

Fault Tolerant Ethernet Overview and Implementation Guide (FE02-300).

Fault Tolerant Ethernet Installation and Service Guide (FE05-300)

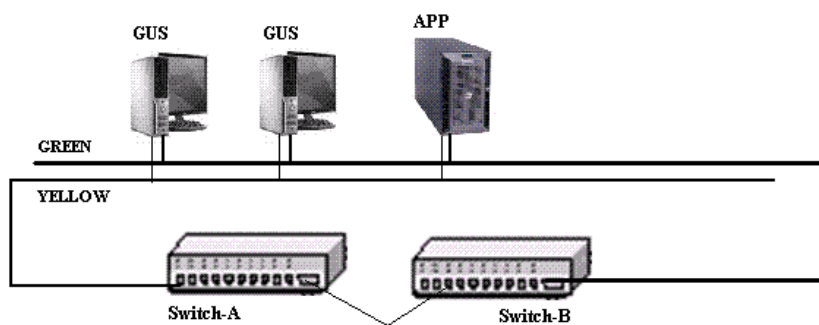
Overview of connecting a node to FTE

FTE connected nodes contain the necessary redundant media components and Honeywell FTE software. FTE nodes connect to the LAN using redundant network interface adapters (each port has a unique IP address). FTE Nodes are flexible to single Ethernet failures such as switch or cables faults. They are able to communicate even if one path exists between them.

Each FTE-connected node has two connections:

- Yellow refers to all the components connected to the primary A switch, each of which is connected using the Honeywell-provided yellow cables.
- Green refers to all the components connected to the secondary B switch, each of which is connected using the Honeywell-provided green cables.

Figure 15 FTE-connected node with two connections

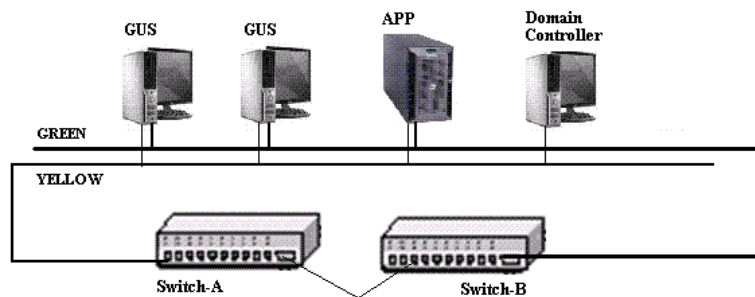


Overview of connecting a Windows 2000 domain controller to FTE**ATTENTION**

- Windows 2000 Server Domain Controllers does not qualify FTE. Therefore, they cannot make both yellow and green connections.
 - Install FTE Driver (from TPS media) on the GUS/APP nodes only.
-

Domain Controller

A Windows 2000 Server Domain Controller connects to an FTE as follows:



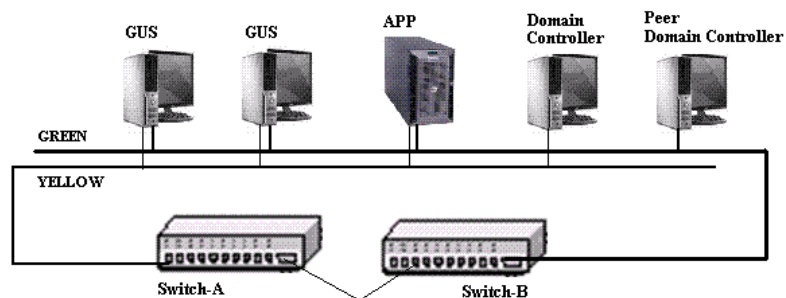
The Domain Controller connects to Yellow.

3 Domain Planning Guidelines

3.10 FTE

Peer Domain Controller

A Windows 2000 Server Peer Domain Controller connects to an FTE as follows:



The Domain Controller connects to Yellow and the Peer Domain Controller connects to Green.

Overview of connecting a Windows 2003 domain controller to an FTE

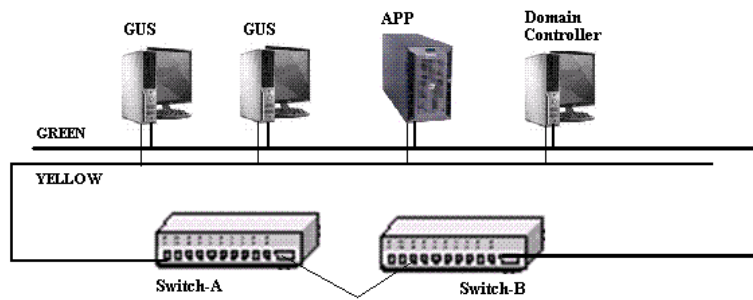


ATTENTION

- FTE is qualified on Windows Server 2003.
 - Install FTE Driver (from TPS media) on the GUS/APP nodes and the Domain Controller.
-

Domain Controller

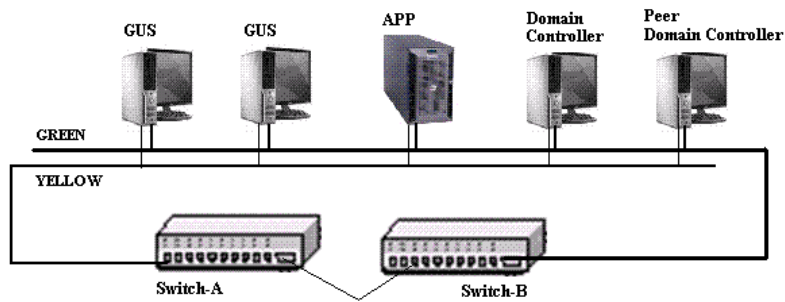
A Windows 2003 Server Domain Controller connects to an FTE as follows:



The Domain Controller connects to both Yellow and Green.

Peer Domain Controller

A Windows 2003 Server Peer Domain Controller connects to an FTE as follows:



The Domain Controller and the Peer Domain Controller connects to both Yellow and Green.

3 Domain Planning Guidelines

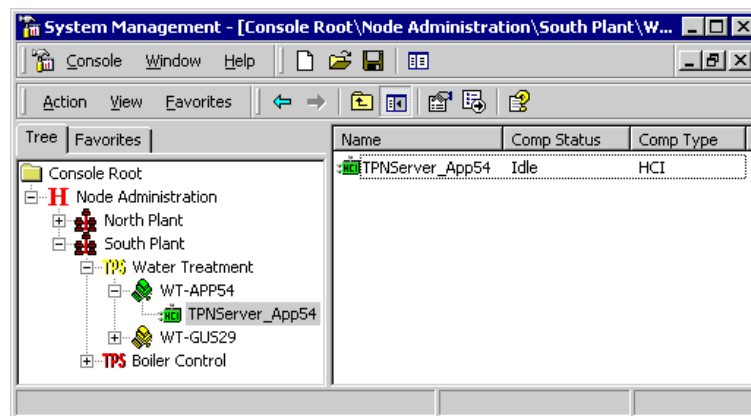
3.11 Windows domain Configuration for TPS

3.11 Windows domain Configuration for TPS

TPS Domain/Console Configuration tool

A TPS Domain or Console is derived from an OUs in the Windows Active Directory. When a TPS Domain or Console is created, it allows you to organize and manage the plant nodes from a System Management Display.

The System Management represents the activity of managing the process control system using the Honeywell system management tool, the System Management Display. The System Management Display provides a hierarchical view of Windows domains, TPS Domain OUs, Computers, and the status of various server applications running in each computer.



The domain or child domain configured into the System Management Display appears under the **Node Administration** root of the System Management view. You can subdivide the domains into number of OUs. The OUs contain the computers. The TPS Domain OUs contain the TPS nodes.

A Domain tree can have number of OUs. A System Management Display can identify and display a TPS Domain OU different from others. The TPS Domain/Console Configuration tool permits you to specify, if an OU is a TPS Domain OU. This specification is included in the System Management Functionality (System Management Display and other System Management Tools) scope. A TPS Console is an OUs within a TPS Domain.

The TPS Domain/Console Configuration tool installs an extension to the Active Directory Users and Computers window. The extension is a TPS Domain tab in the Properties dialog box of an OU.

After installing the TPS Domain/Console Configuration Tool, restart the Domain Controller. The tool installation adds a new tab 'TPS Domain' to the properties of any OU of the domain.

Note: The Microsoft Active Directory schema contains formal definitions of every object class created in an Active Directory forest. The schema contains formal definitions of every attribute that can exist in an Active Directory object. The schema is extensible to define new classes and attributes. Installation of the TPS Domain Configuration Tool (or any other TPS tools) does not change or extend the Active Directory (AD) schema definitions.

Overview of Domain Controller Package

The Domain Controller Package is a software package available on the TPS software media. After installing this package, it builds the Domain Controller part of the TPS Security Model.

When the TPS nodes are included in the domain, the Domain Controller performs the security management.

Action of a Domain Controller Package on a domain

The Domain Controller Package performs the following actions on Domain controller during the Domain Controller Package installation:

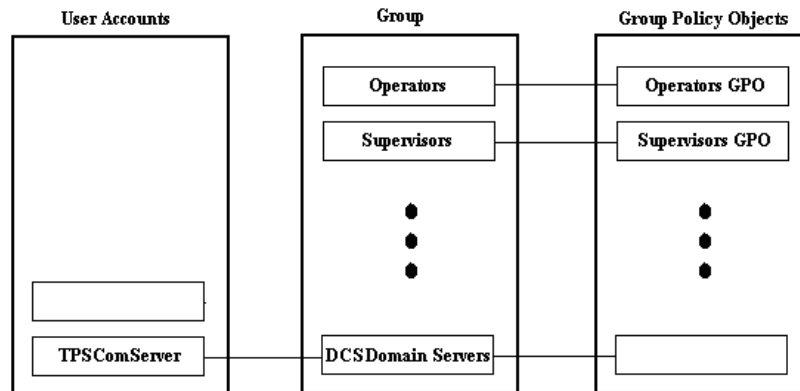
- Creates Group Policy Objects (GPO) for each type of the users
 - Each type of the user has different access privileges. (For example, Operators cannot access Desktop items or Windows Explorer. However, the Engineers can access these items.) GPO maintains these set of privileges for each type of the user. (For example, the operator GPO contains the access privileges of the operators.)
 - The Domain Controller Package creates GPOs for different type of users like Operators, Supervisors, and Engineers.
- Creates global groups and associate it with the respective GPOs.
 - The Domain Controller Package creates the Global User Groups corresponding to every type of users. The corresponding GPOs are assigned to each of the global user groups. These GPOs created have an effect when it is applied to a user group.
- Creates global accounts and assigns the global accounts to the global groups.

3 Domain Planning Guidelines

3.11 Windows domain Configuration for TPS

- The Domain Controller Package creates few user accounts that act as a service account for the COM-based servers. These accounts are assigned to the respective groups.

Figure 16 Domain Controller Package



Overview of a Workstation Package

The Workstation Package is a software package available on the TPS software media. After installing this package, it builds the Workstation part of the TPS Security Model. You must install this package in each TPS nodes of a domain.

Actions of a Workstation Package on TPS nodes

The Workstation Package performs the following actions on the TPS nodes during installation:

- Creates local groups.
- The Workstation Package creates Local Groups corresponding to each type of the users. For example, it creates groups like Local Operators, Local Supervisors, and Local Engineers.
- The access privileges for the local group members are limited. For example, Local Operators have operator level access on the node.
- The workstation package creates user accounts that act as service account for the COM-based servers.
- Provides the LinkDomainGroups.vbs script.

Need for creating local user accounts for users

You must create the local user accounts only for the following TPS nodes:

- TPS nodes that are not USR and AMR
- TPS nodes that not connected to the TPS domain

When a node is not in the domain, then:

- Create a user account corresponding to each user who is using the node.
- Add these user accounts as members of the Local Groups.



TIP

Local user accounts are not required for the TPS GUS and the TPS APP nodes that are in the domain. These nodes use the login accounts from the Domain Controller.

Overview of LinkDomainGroups

The LinkDomainGroups adds all the Global groups (in a Domain Controller) as members of the corresponding Local Groups. This scheme simplifies the security management in the following ways:

- The Domain Controller maintains only a single copy of the user accounts.
- Easy to maintain the centralized information.
- All user accounts have required access on the nodes. For example, an Engineers account in the Domain Controller has an engineer-level access on the TPS node as the Engineers account is a member of the Local Engineers.

3 Domain Planning Guidelines

3.11 Windows domain Configuration for TPS

Consider a network with 10 TPS nodes. The Workstation security is installed in each TPS node. Hence, copy of the Local Groups, Local Operators, Local Supervisors, Local Engineers, and others are saved in every node. Assume there are 2 engineers, 4 supervisors, and 16 operators. If any of these users need to operate any one of the nodes in a domain, the user accounts corresponding to 22 users must be saved in every node. For each node, the following operation is to be performed 10 times:

- Create two user accounts corresponding to engineers and add them as members to Local Engineers.
- Create four user accounts corresponding to supervisors and add them as members to Local Supervisors.
- Create 16 user accounts corresponding to operators and add them as members to Local Operators.

The following are the two major disadvantages:

- It is difficult to create all user accounts on all the nodes
- It is difficult to maintain a consistent copy across all the nodes

To overcome these disadvantages, use the LinkDomainGroups.

When to perform LinkDomainGroups



ATTENTION

Perform the LinkDomainGroups after installing the Workstation Security package and before installing the TPS packages.

The Workstation Security package installs a script file LinkDomainGroups.vbs in \ProgramFiles\Honeywell\WKSecurity path. To perform the LinkDomainGroups, double-click and run the script file.

Overview of LockDownLocalUsers

The policy rights/limitations corresponding to the Engineers, Operators, and other user types are in effect only when:

- A TPS node is in the Windows domain.
- The LinkDomainGroups is performed.

The LockDownLocalUsers is used for building a LOCAL security policy for a node. This node need not be a member of a Windows domain configured with an active group policy in its active directory. After applying the local policy, excluding the local administrator, all users are in a restricted environment. When there is a requirement to change the policy, ensure that the administrator access settings are not changed.

Need for HOSTS and LMHOSTS files in TPS domain

Windows NT uses the HOSTS and LMHOSTS files. The TPS domain uses the Windows 2000 Server and Windows Server 2003 Domain Controllers. These controllers have DNS installed on them and they support WINS.

The peer domain controller in a domain does not require the HOSTS and LMHOSTS. If there is no peer domain controller, these files provide a backup for the name resolution (same as provided by the DNS server). This ensures the TPS software reliability in a failed network.



ATTENTION

The HOSTS and LMHOSTS files must be correct and consistent on every node. For more information, see "COPY_UTILITY.BAT".

Copy_Utility.bat file

The COPY_UTILITY.BAT is a batch file. This file distributes an identified set of files from Domain Controller to an identified set of member computers. The identified files are as follows:

- OPERATOR.BAT file
- HOSTS file
- LMHOSTS file

3.12 Overview of other Windows domain operations

HOSTS and LMHOSTS files

The HOSTS and LMHOSTS files are used in Windows NT.

HOSTS

The TCP/IP utilities (like PING, FTP, and TELNET) use the HOSTS files. These files are saved in the \WINDOWS\system32\drivers\etc path. The following is the file format:

IP Address	Friendly Name
130.25.25.87	GUS30 #Remarks are denoted with a #.

The preceding entry in a HOSTS file permits you to communicate with a specific computer using PING GUS30 and not PING 130.25.25.87. As this file is not dynamic, you must manually add the entries.

LMHOSTS

The LMHOSTS file is used for resolving the LAN Manager name with the TCP/IP protocol. This file resolves the IP addresses for a server that is not in the local subnet. These files are saved in the \WINDOWS\system32\drivers\etc path.

For example, if APP20 is in a different subnet and you type:

NET VIEW \\APP20

The **Server not Found** error message appears. Therefore, to access a computer that is on a different subnet, edit the LMHOSTS file as follows:

IP Address	Friendly Name
130.25.25.87	APP20

Overview of Lookup Zones

The DNS domain zones are database files that contain resource records for a domain, or a set of domains. The following are the two types of DNS zones:

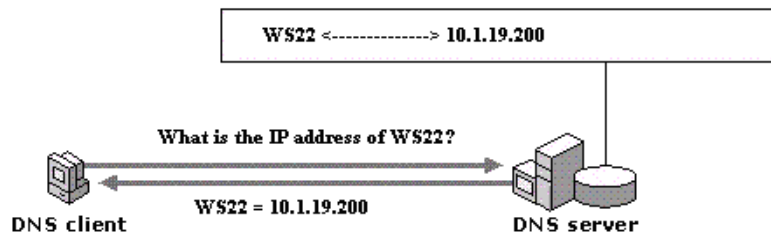
- Forward Lookup Zones
- Reverse Lookup Zones

Forward Lookup Zones

The Forward Lookup DNS zones allow a resolver (an application included in web browsers and most FTP software) to obtain the IP address corresponding to the known host name.

The following figure shows how a resolver obtains the IP address for a known host name:

Figure 17 Forward Lookup Zone

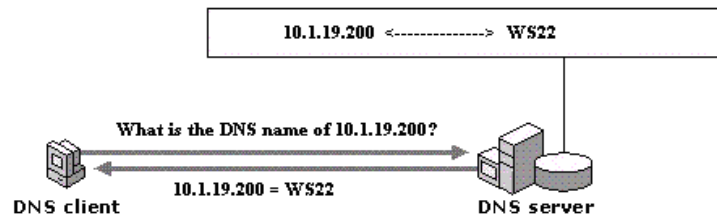


Reverse Lookup Zones

The Reverse Lookup DNS zone allows a resolver to obtain the host name corresponding to the known IP address.

The following figure shows how a resolver obtains the host name for a known IP address:

Figure 18 Reverse Lookup Zone



3 Domain Planning Guidelines

3.12 Overview of other Windows domain operations

Forward Lookup zone files are not configured to respond to the reverse lookup query. You must create a new Reverse Lookup Zone to answer the reverse lookup query.

The Active Directory Installation wizard does not add a Reverse Lookup Zone, because the Reverse Lookup Zone might be controlled by a parent server. You must add the Reverse Lookup Zone to the specific server, if no other server controls the Reverse Lookup Zone. You must add the Reverse Lookup Zone for the hosts listed in the Forward Lookup Zone. The Reverse lookup zones are not required for the Active Directory, but the clients need them to resolve the host names from the IP addresses.

4. Migration Planning from Windows 2000 to Windows 2003

4.1 Windows 2000 Root Domain Controller to Windows 2003 Root Domain Controller

Prerequisite – Windows 2003 server

To perform this upgrade, you will need a retail copy of the Windows 2003 Server Standard Edition CD.



ATTENTION

A root domain controller requires the existence of a Dynamic DNS server. The Microsoft upgrade procedure will offer to configure a Dynamic DNS server on the first domain controller in the new domain.

Honeywell recommends that unless a Dynamic DNS server already exists on the network, accept the auto configuration during the domain controller installation process.

Also, Honeywell recommends implementation of a root domain controller as described in the TPS System Implementation Guide.



REFERENCE - EXTERNAL

Always refer to applicable Software Change Notices for additional details when planning a migration.

4 Migration Planning from Windows 2000 to Windows 2003

4.1 Windows 2000 Root Domain Controller to Windows 2003 Root Domain Controller

Root domain naming must adhere to naming conventions

Enter full DNS name for the root domain.

Note: The DNS name must adhere to DNS conventions (for example, underscores are not permitted). Ideally, the leftmost part of the domain name should reflect the name of your current Windows 2000 domain.

In general, a root domain contains a .local suffix.

Refer to other sections in this guide for additional information regarding DNS naming.

Dynamic DNS server required

Windows 2003 requires a dynamic DNS server (Dynamic DNS is not supported by Windows NT 4.0 DNS services. Windows 2000 supports dynamic DNS Server). As a result, you must configure a Dynamic DNS if one is not already established in the enterprise. The Windows 2003 Dynamic DNS can subsequently be directed to resolve Windows 2003 names and redirect to a Windows 2000-based DNS to resolve other names within the enterprise.

Refer to other sections in this guide for additional details.

4.2 Windows 2000 Child Domain to Windows 2003 Child Domain Controller

Rationale for using child domain controller

A child domain controller for process control is typically established when an existing Windows 2003 domain infrastructure is already in place within the IT enterprise and the process control personnel have fully considered the implications, such as security and Active Directory replication.



ATTENTION

Honeywell recommends implementation of a Root Domain controller as described in the *TPS System Implementation Guide*.

Review the implications of child domains

Administrators should be very familiar with the implications of establishing a process control domain as a child of the IT domain architecture. In addition, the IT and Process Control departments should spend time planning the best possible approach, as the current version of Windows 2003 does not offer tools to modify the domain architecture after it has been established.

Refer to other sections in this guide for additional Honeywell guidelines on Windows 2003 domain architectures.



REFERENCE - EXTERNAL

Always refer to the Software Change Notice for additional details when planning a migration.

4 Migration Planning from Windows 2000 to Windows 2003

4.2 Windows 2000 Child Domain to Windows 2003 Child Domain Controller

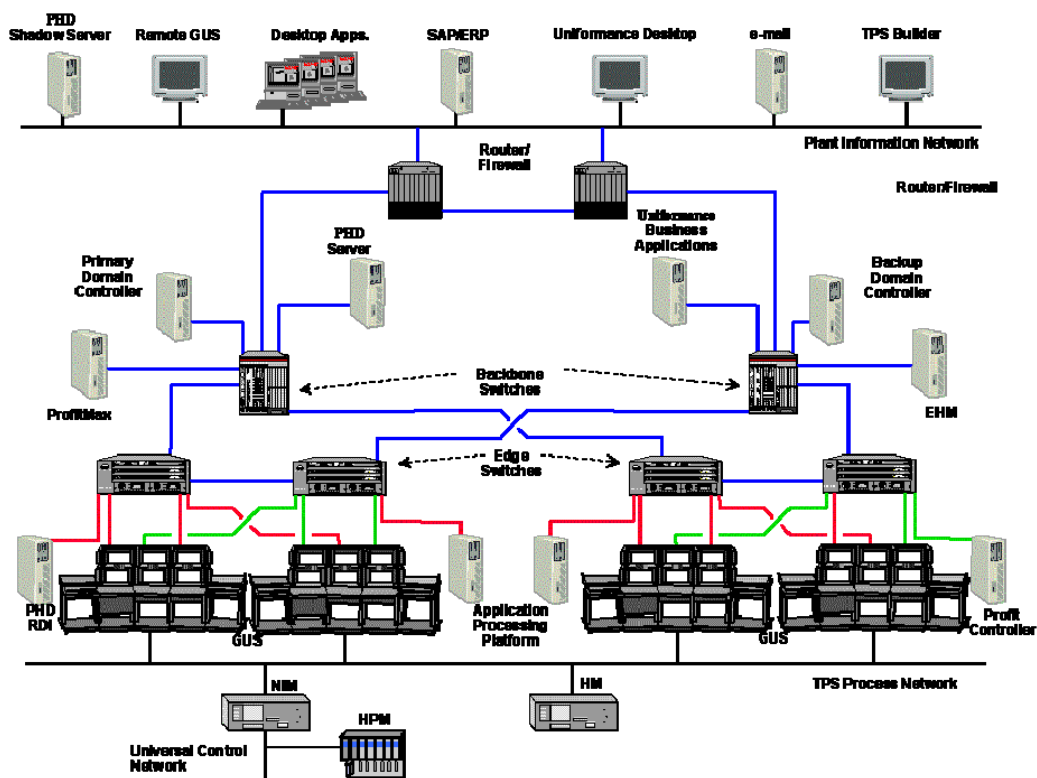
5. Industrial Strength Networking Concepts

5.1 Concept Overview

Redundant topology

When there is a requirement for the installed system to connect to a network that delivers the highest levels of reliability and availability, Honeywell recommends the physical architecture shown in the following diagram:

Figure 19 Industrial Strength Networking



5 Industrial Strength Networking Concepts

5.1 Concept Overview

In the recommended arrangement, TPS domain nodes and non-TPS domain nodes connect to the ISN backbone through a redundant network topology. This type of connection has the following advantages:

- The network elements that provide redundancy (for example, an alternate data path, or an alternate active device such as a server, switch, or router) do not need to be co-located with the primary network elements. This reduces the probability that problems with the physical environment (such as power or cable break) will interrupt service.
- Problems (such as communications errors/failures, system bugs/upgrades, and device configuration faults/changes) can be dealt with separately, and more importantly, without disruption to service.
- With redundancy provided by the network topology, each network device no longer needs to be configured for the ultimate in standalone fault tolerance, thereby reducing the cost of each network device and the overall cost of computing.
- The traffic load between the respective layers of the network topology (TPN, and ISN) can be shared between the primary and alternate data paths – this increases the aggregate network performance and capacity.
- The ISN topology can be configured to automatically fail-over from the primary to secondary paths without operator intervention.

Honeywell Network Technology support

The Honeywell Network Technology Group can assist customers with planning, design, and implementation of Industrial Strength Network systems. This support includes TPS systems, Experion systems, and TPS systems that are integrated with Experion.

Secure PIN to PCN connectivity

At a minimum, a router should be used to isolate the ISN from the Plant Information Network. The sub-netting of the control network will accomplish the following:

- Eliminate unnecessary traffic like broadcasts and errors on the control network.
- Provide packet filtering of incoming and outgoing data in the control environment.
- Provide encryption of sensitive information.

A router provides the first line of security. It is highly recommended that a “Firewall” be established between the business information and the control system environments. This firewall (a component or set of components that restricts access between a set of networks) can be programmed to enforce specific policies. Firewalls provide security on the application (stateful or stateless inspection), protocol, and/or addressing level; they also can build-in encryption keys and higher levels of security than a router. The use of a firewall dramatically improves the security of a private network from external threats.



ATTENTION

Honeywell recommends implementation of a Stateful and/or Stateless Inspection Firewall.

Stateful or Stateless Inspection Firewalls check every packet on the network at all levels of the OSI model. A Stateful and/or Stateless Inspection Firewall has the following features:

- security
- performance
- application-layer awareness
- scalability
- extensibility
- transparency

Stateful and/or Stateless Inspection Firewalls are the most cost effective and the most secure type of firewall.

5 Industrial Strength Networking Concepts

5.1 Concept Overview

6. Security Planning for TPS Systems

6.1 Overview of Security Planning

Normal security considerations apply

After the PCN is connected to a PIN (particularly a PIN that uses WAN, intranet, or even internet access), all of the normal networking considerations apply with respect to user rights and security. In this respect, a TPS node is no different than a networked PC workstation on a LAN.

Example security considerations

The following are a few of the topics customers need to consider PIN to PCN Access Rights:

- Remote Access Security
- Anti-virus measures
- E-mail Implications
- WAN Access and Security
- Intranet Access and Security

Coordination between IT and process control is critical

The previous topics are generally well understood by the customer's own Networking or Information Technology staff; however, the TPS system comes with a "standard" security configuration, which may differ from that of the IT department. Therefore, coordination between IT and Process Control environments is critical to a successful implementation. Honeywell Services can provide a full range of network services, including consultation, network setup, remote monitoring and system administration.

6.2 TPS System Security Considerations

TPS system security configuration philosophy

The TPS system security philosophy should provide every user with the minimum rights and functionality required to effectively perform their duties. Where there is a PCN, the philosophy is deliberately to set up the PCN as a “closed” network. This means that the various PCN nodes will have sufficient functionality to meet the needs of the overall PCN but, in general, will not include the more wide-ranging rights and functionality required for a general Plant Information Network PIN. This approach makes it easier to keep the PCN secure. When additional rights and functionality are required to meet the more general needs of the PIN and its users, adding these becomes a conscious, deliberate decision and details of the required changes can be designed as the alternative to the precise, application-specific needs of users. This approach is consistent with normal networking practices and follows closely the high security recommendations from Microsoft.

Default TPS system security capabilities

To provide adequate rights to each user to perform their tasks, the general functionality given to each user level is as follows:

Table 9 Default TPS System Security Capabilities

User Type	Default TPS System Security Capability
View only users, Operators, and Supervisors	• Access to LCN, graphics, and SafeView • Access to system status through MMC snap-in and selected applications only • Ability to shut down the node • View only users are restricted from writing TPN data • No access to LCNP Status • No access to Windows Explorer • No access to removable media (such as CD-ROM drives) • Password is blank (except for Supervisors where it is required)

User Type	Default TPS System Security Capability
	No screensaver allowed
Engineers	- Access to LCN, graphics, SafeView, and all engineering tools - Access to TPS system status display - Write access to limited directories and files - Ability to shut down - Access to removable media - Password and screensaver required
DCS Administrators and Windows 2003/XP Administrators Note: In TPS R400, TPS administrators are changed to DCS administrators	- Unlimited Windows 2003/XP local node access (except TPS Administrators are restricted from registry and Policy Editor functions) - Access to removable media - Password and Screensaver required
Backup Operators	- Limited access only to allow proper operation of backups - Access to removable media - Password and screensaver required

TPS system security enforcement

TPS security is enforced through the combination of File System Security, User Rights, and System Policies. A TPS system also depends on the existence of specific Local Users, Domain Users, and Domain Groups. The next section lists the reasons for TPS security mechanisms and provides recommendations to ensure network security.



REFERENCE - INTERNAL

For details on the specific permissions, rights, and policies for GUS and APP nodes, refer to the *TPS System Administration Guide*.

For LCN-connected Experion nodes (ES-T, ESVT, and ACE-T), refer to the *Experion Network and Security Planning Guide* and the *Experion Server and Client Configuration Guide*.

6.3 General TPS System Security Rationale

Security considerations based on user type

Table 10 Security Considerations Based on User Type

User	Security Consideration
Operations	• The operator should not be required to remember a password to handle emergency situations quickly. Therefore, no password is required. The administrator may optionally choose to set the system up such that the operator login occurs automatically. • A local GUS login is provided to ensure that there is always view to the process for Native Window displays or GUS displays using the GUS Data Server even if all domain controllers are down. • Similarly, a local Experion login is provided to ensure that there is always view to the process for Native Window or Experion displays. • Access to files is restricted by only allowing a small set of applications to be run, rather than by setting no access to the files.
Engineering	• The engineer's write access is restricted to the engineering databases and files such as Experion Display files or TPS Builder database. • Screensavers with passwords are used to secure the desktop from unauthorized access.
Administration	• The administrator should be the only one permitted to change the registry and profiles because this is critical to the operation of the system. • Domain groups are used for easier administration of permissions and user rights. • User rights are configured to only allow TPS system users to access TPS nodes. • Hidden shares are used for distribution of modified files from a central location. • Screensavers with passwords are used to secure the desktop from unauthorized access.

Security recommendations based on component

Table 11 Security Considerations based on Component

Component	Configuration Consideration
Secure TPS Node	• Robust machines (with error checking and correcting memory) • Use Windows 2003 Security and NTFS, have local login available • Removable media locked down • Use auto login on nodes where needed • Do not allow users to install software (only Administrator) • Disable guest account • Use Time-to-Live counters to limit network scope • Provide physical security for Server Nodes
Secure PCN Subnet	• Have a separate IP subnet connecting LCN-connected Experion nodes (ES-T, ESVT, and ACE-T), GUS, APP, and other TPS nodes in Control Room • Use static IP addresses • Limit the domain to nodes in the control room • Do not install FTP, TELNET, RSH, REXEC services • All tools should use Windows 2003 Security • Use managed switches instead of hubs • Use static routes, limit SNMP access to critical performance data • Follow cabling infrastructure standards for reliable and robust cabling
Secure Backbone	• Keep administrator authority in the local domain • Keep user accounts in the local domain (or at least on a backup domain controller) • Connect only to secure nodes • Keep printers in the local domain • Provide reliable name resolution

6 Security Planning for TPS Systems

6.3 General TPS System Security Rationale

Component	Configuration Consideration
	Do not use dynamic routing protocols
PIN to PCN Connectivity	- Keep connections to other networks behind a firewall - Deny inbound access by default - Use only IP protocol - Use static routes on firewall (use IP filtering)

Table 12 Security Settings by Node Type

Node Type	Security Policy	Description
TPS node (includes GUS and APP)	Apply the High Security Policy Note: In TPS R400 TPS Security Policy is replaced with Experion High security Policy	TPS Nodes run the traditional TPS system security tools for applying a locked down configuration.
Domain Controller (Windows 2000 Server and Server 2003)	Apply the High Security Policy	Run the system security tools for applying global groups (Operators, Engineers, and Supervisors) and group policies, which control the desktop environment.
LCN-connected Experion nodes (includes ES-T, ESVT, and ACE-T)	High Security Policy	LCN-connected Experion nodes with TPS options use the High Security Policy for applying a locked down configuration.

7. Performance Planning for a TPS System

7.1 Overview of TPS System Performance Planning

Purpose of performance planning section

This section describes a performance model upon which performance planning and load analysis can be done prior to installation to ensure that there is enough CPU, disk, and data access throughput available to meet the needs of your site. It also describes alternatives that can be taken to reduce the load.

Loading analysis resources

Loading analysis must take into account the following critical resources:

- TPS node CPU availability
- TPS node memory utilization
- TPS node disk capacity and throughput
- PCN communication loading (Ethernet)
- TPS Network data owner loading (NIM/HPM, HG, and so forth)
- TPS Network communications loading (LCN/UCN)
- PHD client loading
- Database capacity and throughput

7.2 Overload Behavior Philosophy

Provide spare capacity

In designing a TPS system, enough spare capacity for the critical resources described previously should be designed to avoid any overload problems that can occur.

Overload behavior

If, however, such an overload does occur, the TPS system software behavior is as follows:

- TPS nodes (GUS, APP) and LCN-connected Experion nodes (ES-T, ESVT, and ACE-T) are self-throttling on their data sources. That is, as client loading increases; response time becomes long and throughput to the clients decreases, but the load on the data source will be limited to a maximum rate.
- As the CPU approaches 100% utilization, or memory use exceeds physical memory capacity, or disk requests reach their capacity, response time becomes longer and throughput decreases, but the TPS system software will continue to execute and ride through the situation.
- When the PCN network loading reaches its saturation point, the response time on communication between nodes will become longer, the throughput between nodes will decrease and messages may be lost. However, local-only operations such as communication to the TPN through an LCNP card should continue with minimal impact on response time for that data.
- When TPN data owner loading reaches its saturation point, data requests for highest priority clients are served first. Response time for lower requests will become longer. If high priority requests exceed the throughput they also will see an increase in their response time.
- When TPN communication loading reaches its saturation point (~70% load and caused by multiple requestors asking for too much data), the following can occur: a redundant node fails, response time becomes long, AM overruns occur, HM history overruns occur, and LCN cable failures.
- As database throughput becomes heavily loaded, client response time becomes longer and total throughput decreases, but requests continue to be satisfied in the order received.

7.3 TPN Loading Analysis

Determining bandwidth

The LCN-connected Experion nodes (EST, ESVT, ACE-T), and the GUS, HM, PHD, AM, and APP all put load on the TPN system. In a system with one or two clusters, the first concern is with the load on the NIM. Additional NIMs may be added to solve overloaded Network Interface Manager (NIM) problems. Where the system has more than one or two clusters, the next concern is with LCN bandwidth. To determine if there is a bandwidth problem, add up the loads from the nodes listed above.

History load

The HM is limited to 40 parameters per second (2400 parameters per minute). PHD has the capability to collect a large number of points per second. Therefore the load on TPN must be incorporated into performance analysis. For example, if 12000 points were being collected on a 1-minute collection rate, that would give a load of 200 parameters per second on TPN.

For systems with an Experion Server TPS (ESVT), the primary history collector is the Server. HM history is required only to support legacy displays such as Native Window historical trends. However, the above analysis approach is still valid.

Advanced control load

TPS system applications load must also be included in the load on the TPN. For example, 500 parameters gathered every 5 seconds adds an average load of 100 parameters per second to the TPN.

AM Control load must be included for each AM and APP on the system. If it is executing 50 loops per second, with each loop requiring an input and an output, the parameter load will be 150 parameters per second.

Human interface load

In a maximum load situation, a Experion Station TPS (ES-T) or GUS with a single display of 400 parameters all going to the TPN network and updating on a 4-second update cycle would generate a load of 100 parameters per second on the TPN. Assuming also that each Experion Station TPS (ES-T) or GUS has five displays (including a Native Window graphic), and then 16 Experion Station TPS (ES-T) or GUS on a system would give a total potential load of 8000 parameters per second. This far exceeds the data bandwidth of the TPN, which is 4000 parameters per second.

7 Performance Planning for a TPS System

7.3 TPN Loading Analysis

The actual load on the total number of parameters being collected and the total number of displays is a crucial factor in determining if enough bandwidth is available. Ensure that the number of parameters is correctly estimated. Each PV and its alarm status require 2 parameters to change color.

When a Server is present, it may also represent an additional human interface load, since it may be supporting ES-Fs and remote operator stations via DSA. In this case, the additional parameter load represented by these displays must also be included in your load calculations. You may also consider whether or not these displays are accessing the same parameters that are being historized by the Server. If so, the LCN accesses are consolidated, and this should be factored into your estimation.

Some displays may be primarily PHD data that does not load the TPN. In addition, other stations may be used only for configuration. Be sure to include this type of load in your PCN load estimation.

NIM/HG load assessment

Each K4LCN NIM can handle up to 8000 tags at 2400 parameters/second. Each HG can handle 3000 parameters at 1100 parameters per second. In most cases, however, the load is much less than this. Therefore, the amount of load on each data owner should be calculated based on the worst case number of data requestors for data through that node (include GUS, AM, HM, and so forth).

Total TPN load assessment

After careful determination of each of the previous loads, add them up to determine if a loading problem exists. The cluster guidelines defined in the *TPN Engineer's Reference Manual* include the following:

- The total TPN loading is the sum of all the loads described previously.
- The TPN can handle 4000 parameters per second.

If your assessed load is close to 4000, use the alternative listed next.

7.4 Alternatives for TPN Load Reduction

General load reduction

General Load Reduction	✓
Design your use of multiple displays to optimize the load on the TPN Servers.	
Reduce the update frequencies for non-critical control data on displays.	
Reduce the collection frequency for history.	
Follow the HMIWeb Display Builder and the GUS Display Builder guidelines for developing graphics. REFERENCE: For optimizing HMIWeb display performance, see "Using the Display Performance Analysis Tool" and "Improving the Performance of an Existing Display (an Example)" in the <i>HMIWeb Display Building Guide</i>.	
Use Single Scan Per Data Item option (the default option). REFERENCE: For more information, refer to "Performance Considerations" in the <i>TPN Server User's Guide for Windows 2003/XP</i>.	

LCN load reduction

LCN Load Reduction	✓
Use HCI/OPC server caching as much as possible (TPN Server, PHD OPC Server).	
Follow the Honeywell OPC performance guidelines. REFERENCE: For more information, refer to "OPC Performance Guidelines" in the <i>HCI Client Developer's Guide</i>	

NIM load reduction

NIM Load Reduction	✓
Add additional NIMs (if the total TPN load is not already consumed).	
Ensure that the NIM is using K4LCN board.	

7 Performance Planning for a TPS System

7.4 Alternatives for TPN Load Reduction

8. Non-TPN-Connected Node Planning

8.1 Installation Requirements for Non-TPN-Connected Nodes

System requirements

The following table lists the system requirements to run TPS system software on a node that is not connected to the TPN/LCN (the node may be part of the TPS domain):

Item	Description
Processor	Pentium Pro 200 MHz or greater.
Memory	128 MB minimum.
Hard disk	Always allow at least 20% free space. In addition, any database requirements must be considered.
Operating System	Windows 2000 or Windows 2003 or Windows XP.



REFERENCE - INTERNAL

For installation details, refer to the *TPS System Implementation Guide*.

For installation requirements of Experion nodes, refer to the *Experion Software Installation and Upgrade Guide*.

Software requirements

If you want to perform application development on a non-TPN-connected node, then install Visual Studio SP4 (contains both Visual C++ and Visual Basic).

You may install many application packages on a non-TPN-connected TPS node, and you need to do capacity planning for these applications. See the respective packages for their capacity requirements.

8.2 Planning TPS Access from a Non-TPS Node

Non-TPS node definition

A non-TPS node is a node that resides outside of a TPS domain, but is a member of a Windows 2000 or Windows 2003 domain.

Planning objective for non-TPS node access

The following procedure will allow HCI client applications on a non-TPS node to communicate with a server residing within a TPS domain. The TPS server can be a TPN Server on an APP or on an Experion Server-TPS (ESVT).



ATTENTION

Typically, one-way trust relationships restrict asynchronous DCOM callbacks. This restriction limits OPC communication to synchronous calls (polling) only. A two-way trust relationship (including a Windows 2000/2003 transitive two-way trust) eliminates the callback restrictions.

Planning goal for non-TPS node access

The non-TPS node and its PDC/Domain Controller, and the TPS domain node and its Domain Controller must all be visible to each other over the Ethernet network.

Establish trust relationships to support OPC communication

Unique configuration challenges emerge when clients in one Windows 2003 or Windows 2000 domain attempt to communicate with servers in another domain. Clients using DCOM (this includes OPC clients) as a communication mechanism is particularly impacted.

Windows 2000/2003 simplifies configuration problems by automatically establishing transitive two-way trust relationships for all domains within a forest. If the domains are not members of a common forest, users can still establish one or two way legacy Windows 2000 trust relationships between domains (Windows 2003 to Windows 2000 or Windows 2003 to Windows 2003 in another forest) to achieve the desired functionality.

Add security after trusts in place

After appropriate trust relationships are in place, additional security configuration is recommended. The TPS system applies security to numerous files, registry keys, and user rights to provide a highly secure computing environment. All security is applied using local Windows XP/2003/2000 groups. To operate correctly in a domain environment, global groups are added to local groups on each node. The default High Security Policy installation performs these steps automatically.

If you want users in other domains to be able to access local resources (servers or the registry, for example), add the appropriate global groups or users from the remote domain into the local groups on each machine.

Example:

- Local group, DCS Engineers, is applied to the User Right: **Access this computer from the network**.
- The domain Engineers group in ProcessControlDomain1 is automatically added to the local DCS Engineers group on each node. After doing so, users who are members of the ProcessControl Engineers global group are able to access each TPS node from the network.
- Users from BusinessDomain1, however, are unable to access each TPS node. To correct this, create an Engineers global group on BusinessDomain1 and add this new group to each of the local TPS Engineers groups.



REFERENCE - INTERNAL

Refer to the *TPS System Administration Guide* for additional information.



ATTENTION

Windows XP SP2 and Windows Server 2003 SP1 provide new security measures which include enhanced Remote Procedure Call (RPC), DCOM, and Windows Firewall settings. Under Windows XP SP2 and Windows Server 2003 SP1 the host-based firewall will, by default, reject any incoming connections; however, installing TPS Software will enable the Host firewall for secure access to the Server.

8 Non-TPN-Connected Node Planning

8.2 Planning TPS Access from a Non-TPS Node

9. PHD Considerations

9.1 Overview

PHD planning considerations

Process History Database (PHD) planning includes where to locate the history database, where to locate the relational database, considerations for multi-platform issues if not all nodes are Windows-based, defining what to assign to history, whether or not shadow PHD servers are needed, and where to locate Uniformance desktop clients. These considerations and how to plan for them are addressed in the Uniformance documentation set, and in training available through Honeywell Automation College. Additionally, help in understanding and implementing PHD can be obtained through Honeywell Services. This document does, however, cover some PHD considerations such as:

- Things to consider when selecting a PHD data collection interface.
- Security considerations when using PHD OPC Server and PHD.

PHD provides a variety of interfaces for data collection from Honeywell and third party data sources.

9.2 Integration of PHD with Experion and TPS Systems

Data collection mechanisms

To historize TPS and Experion data into PHD, the user can choose between the following data collection mechanisms:

- Collect data using a **Real-time Data Interface (RDI)**.
- Obtain data already cached in the Experion history database using the **Experion Link** (Requires PHD R210 or later and Experion R300 or later).

Integrate PHD with existing TPS system

PHD may collect TPS data through one or more Real-time Data Interfaces that run on an APP node. The following RDIs may be used, depending on site requirements:

PHD Interface	Description
LxS RDI	The RDI software must be installed on the APP node. The PHD Server software must be installed on the APP node.
OPC RDI	The RDI software and PHD Server software must be installed on the APP node. To access TPS data, the RDI is used in conjunction with the TPN Server.

Note: Any user upgrading their TPS system to Experion may continue to use PHD integration through the APP node.

Integrate Experion and TPS systems with an added PHD

If you require PHD functionality as you integrate Experion and TPS systems, then you may deploy an APP node-based PHD as stated previously, or you may choose to use the Experion Link for Experion R3xx and PHD R210 integration.

Figure 20 PHD Interfaces for APP Node

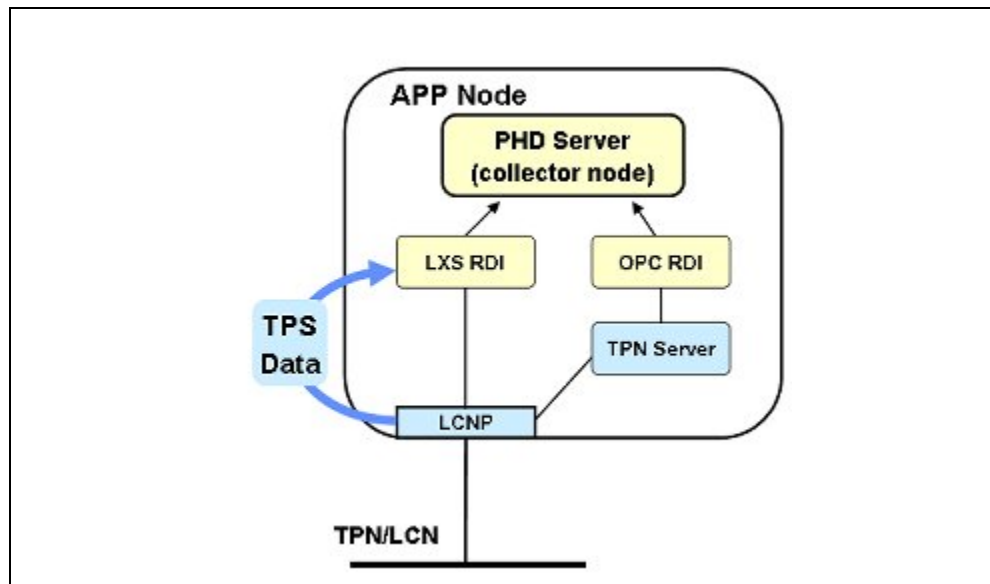
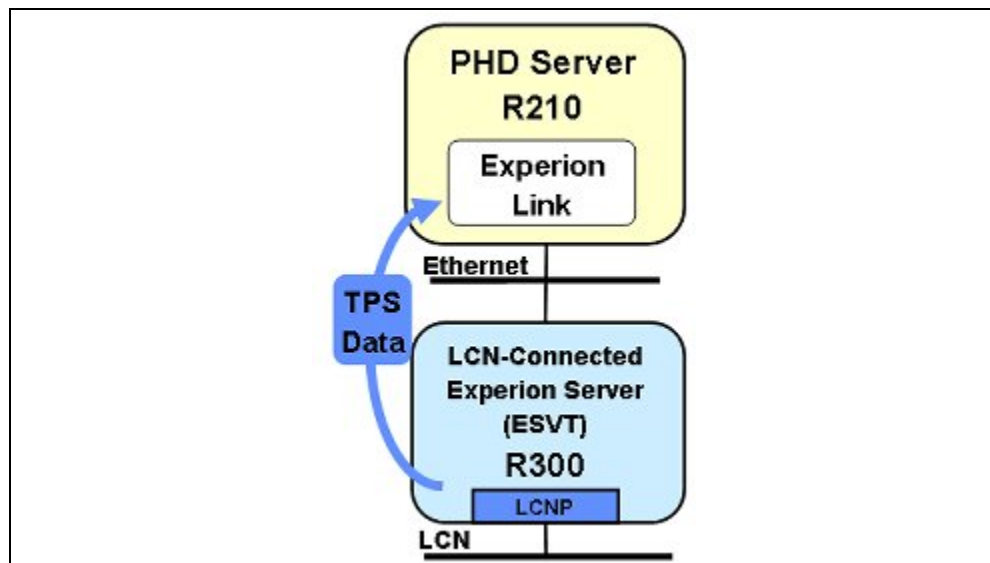


Figure 21 PHD Interfaces for LCN-Connected Experion Server



9.3 Overview of PHD Interfaces for TPS Systems

APP node - Direct Connection – LXS Real-Time Data Interface

The TPN/ LXS RDI (Real-Time Data Interface) is a polled interface using the Honeywell LXS Data Access API. The RDI can access all TPN/LCN tag parameters, including Custom Data Segments (CDSs).

Features of the LxS RDI are the following:

- Ability to access all TPN data types including Custom Data Segments.
- Ability to more fully control data collection load and LxS channel use through RDI configuration when PHD is the sole application utilizing the APP node.
- Note that if multiple applications use the APP node, there is no protection from duplicate data collection by various applications.



REFERENCE - INTERNAL

For more information on the TPN/LXS interface, refer to the *TPN/LxS RDI Installation Guide* in the PHD documentation set.

APP node - TPN Server – OPC Real-Time Data Interface

The OPC RDI (Real-Time Data Interface) uses the OPC DA (Data Access) standard for normal data collection. The OPC RDI supports polled data collection or subscription-based exception data collection.

Features of the OPC RDI when used with TPN Server are the following:

- Access to TPN data types (slightly more limited access than with LxS RDI)
- Ability to run in exception mode to reduce system loading.
- Accesses the TPN Server's data cache, thus reducing system load when multiple applications make data requests of the same APP node's TPN Server.
- Uses the OPC standard to ensure that the PHD application is isolated from any changes to the basic operation of the APP node.
- Uses the OPC standard to make future transition to new systems, such as Experion, simpler than with a TPN-specific interface.

**REFERENCE - INTERNAL**

For more information about the OPC interface, refer to the *OLE for Process Control RDI Installation Guide* in the PHD documentation set.

9.4 Overview of PHD Interfaces for Experion Systems

PHD/Experion Link between Experion R3xx and PHD R210

Some TPS sites can choose to have PHD (running PHD R210) collect TPS data through an LCN-Connected Experion Server (ESVT) running Experion R3xx. This does not mean every system using an APP node to collect TPS data for PHD can migrate to an LCN-Connected Experion Server.

The Experion Link is the data transfer component for the integration of Experion R3xx and PHD R210. The additional capabilities of this integration include automatic tag synchronization and inclusion of PHD data on Experion trends.

Features of the Experion Link are the following:

- Tag configuration may be automatically maintained using Experion Tag Synchronization.
- No duplication of data collection. Data is collected by Experion history, and PHD does not generate added load.
- One interface per Experion server. There is no need to perform load balancing based on collection volume, collection rate, or availability of history recovery.

**REFERENCE - INTERNAL**

For more information on the Experion Link, refer to the *PHD/Experion Link Installation and Configuration Guide* in the PHD documentation set.

PlantScape RDI for access to Experion C200-based data

PHD can collect data from Experion C200-based systems using an established PlantScape/Experion Real-Time Data Interface (RDI) or the Experion Link.

Both the RDIs and the Experion Link allow users to layer PHD above an Experion system, provide data collection from redundant Experion servers, and recover history from the Experion history archives.

9 PHD Considerations

9.4 Overview of PHD Interfaces for Experion Systems

The PlantScape RDI (Real-Time Data Interface) works with the local or network API on Experion R1xx, R2xx, R3xx, and PlantScape R3xx, R400, and R500.

Features of the Experion/PlantScape RDI when used with TPN Server are the following:

- Supports polled and exception-based data collection.
- Supports history recovery when the tags are defined in Experion history.



ATTENTION

In the current releases of PHD and Experion, the PlantScape/Experion RDI should not be used for the collection of LCN-based data.



REFERENCE - INTERNAL

For more information on the PlantScape interface, refer to the *Experion/PlantScape RDI Installation Guide* in the PHD documentation set.

9.5 Performance, Capacity, and Features of APP and ESVT for PHD

Server nodes

- Maximum number of nodes that can connect to one LCN:
 - APP: 20
 - ESVT: 1 (optionally redundant)
- Maximum data collection rate per node:
 - APP: 350 parameters per second
 - ESVT: 1200 parameters per second (capacity for *all* users of LCN data on the ESVT). For optimal throughput, all numeric data collected by PHD should also be assigned to Experion history.
- The amount of data that can be collected from one LCN using multiple APP nodes is greater than can be collected by a single ESVT.
- Maximum number of servers that one PHD Server can connect to:
 - APP: No fixed limit, determined by load
 - Experion Servers: 3
Sites that require connectivity to more than three Experion Servers will need to add PHD Server collectors.

PHD collection rate

With an APP node, the PHD tag configuration determines the PHD collection rate. You can use any rate (1 second or greater) to collect any LCN parameter.

With the LCN-Connected Experion Server and Experion Link, synchronized PHD tags are collected from the LCN at the rate determined by the Experion history type (Fast, Standard, or Extended). The setting of the PHD tag does not determine the collection rate.

LCN Event Journal

Sites that require TPS Event Journal Collection (EJC) require an APP node. Event Journal Collection (EJC) transfers TPS events to an Oracle database. EJC is not supported on an ESVT.

9 PHD Considerations

9.5 Performance, Capacity, and Features of APP and ESVT for PHD

An LCN-connected Experion Server does not support storing string or time data types in history; consequently, history recovery of these data types to PHD is not possible, because they are not stored in history on the LCN-Connected Experion Server.

Redundancy

On systems using APP nodes and Robust Data Collection for redundancy when a collector is shut down and the standby node takes over, there is very little loss of history on PHD.

With redundant LCN-Connected Experion Servers, when one server is shut down, there will be a small loss of history on PHD (potentially, up to one minute).

Firewall support

The Experion Link uses OPC for data transfer to PHD - for most users, OPC is not acceptable for use through firewalls. In this situation, you may require an additional PHD collector inside the firewall to be able to use PHD to PHD communication through the firewall.

Summary of considerations for APP node replacement

Consideration	APP Node (Using TDC_LxS or OPC RDI)	LCN-Connected Experion Server (Using Experion Link)
Performance	350 pps maximum 20 per LCN No limit per PHD Server	1200 pps maximum (impacted by other TPS data users) 1 per LCN Maximum 3 per PHD Server
Event Journal Collection	EJC supported	EJC not supported
History Recovery of String and Time data types	Available	Not Available
History Collection Rates	Any user defined collection rate	Predefined collection rates only: Fast: (1 to 30 seconds) Standard: 1 minute Extended: 1 hour

Consideration	APP Node (Using TDC_LxS or OPC RDI)	LCN-Connected Experion Server (Using Experion Link)
Tag configuration and maintenance	Manual or semi-automated through PHD's tag loader.	All Experion history tags may be automatically created and maintained in PHD using Tag Synchronization.
Redundancy	Robust Data Collection – minimal loss of history on node shutdown	Experion Redundancy – up to 1 minute of history loss on server failover
Firewall Support	Collector to Shadow communications using a small number of user definable TCP ports	Uses OPC, most likely require additional PHD Collector

9.6 PHD Security

Overview - PHD security

PHD is a heterogeneous product that runs on multiple platforms and, of necessity, implements a security scheme that is portable across multiple operating systems. This means that the security configuration is not dependent on a specific operating system registry.

PHD security is used by numerous Uniformance applications. These applications can be integrated with the TPS system, but may be installed as independent applications on a TPS node.



REFERENCE - INTERNAL

For information on the PHD security features, see the *PHD Security and Network Planning Guide* in the PHD documentation set.

9 PHD Considerations

9.6 PHD Security

10. Third Party Component Planning

10.1 Third Party Component Policy

General considerations when using any third-party component

As the TPS system is used to control processes, caution must be used when adding third-party hardware or software components. The installation and addition of third-party software or hardware components could introduce compatibility issues, and thus affect the stability of the system and the view to the process.

Therefore, this section provides Honeywell's policy for adding third-party components to the TPS system, the Portfolio Management qualification initiative and process, and guidelines to use if a third-party component is being considered.

Third-party software compatibility policy

Honeywell's policy for the use of third-party software components on any TPS system product where Process Solutions does not provide the component is as follows:

Any third-party software component must be tested to be compatible prior to its installation and operation on any TPS node. The customer assumes all responsibility for issues resulting from the installation and operation of components that have not been tested as compatible by Honeywell. In addition, it is the customer's responsibility to ensure that the combinations of third-party components have sufficient resources available for proper operation (for example, memory, CPU performance, disk space, display space, and so forth).

Third-party hardware compatibility policy

Honeywell's policy for the use of third-party hardware components on any TPS system product where Process Solutions does not provide the component is as follows:

Any third-party hardware component must be tested to be compatible prior to its installation and operation on any TPS node. The customer assumes all responsibility for issues resulting from the installation and operation of components that have not been tested as compatible by Honeywell. In addition, hardware components that require incursion into the TPS node hardware chassis must be certified and installed by

10 Third Party Component Planning

10.1 Third Party Component Policy

Honeywell. The installation of a component inside a TPS node chassis by the customer voids the warranty on that TPS node.

Third-party component testing

Honeywell maintains a list of third-party components that are qualified for use on the TPS system. Honeywell's Portfolio Management initiative will qualify many of the more generally known and more frequently used/requested third-party software components. Honeywell cannot, nor will ever be able to qualify all possible third-party components. However, a service is provided whereby Honeywell will qualify customer-specific third-party software or hardware components on a fee basis. If desired, a customer can perform their own compatibility testing; however the customers assume all responsibility for the result.

Honeywell's compatibility testing is focused on the interaction and impact assessment of the installation and operation of the third-party component, and does not address the functionality or interaction (such as the exchange, embedding or sharing of data or objects) between third-party components and TPS system products. This compatibility testing is to ensure that the third-party component does not have an adverse side effect or impact on the operation of the TPS system.

In addition, not every combination of third-party components is tested for compatibility. As indicated in the policy, it is the customer's responsibility to ensure that desired combinations of third-party components have sufficient computer resources available (memory, CPU, disk space, and so forth).

Third-party OPC compliance test policy

The HCI subsystem is specifically designed to implement all required interfaces of the OPC specification. To ensure that these interfaces function properly, and to verify that no Honeywell-specific interfaces are required for OPC clients and servers to connect to Honeywell clients and servers, all Honeywell-developed HCI-based components will be tested against third-party OPC components.

Testing of Honeywell HCI components for OPC compliance will be done through the use of the compliance test suite provided by the OPC Foundation's Compliance Subcommittee. Compliance testing will be conducted on each major release of each component. In addition, minor releases will be compliance tested if they are specifically expected to address compliance issues. Each component will be tested to the highest release of the OPC spec with which it is stated to be compliant.

In the absence of a formally accepted OPC Foundation Compliance Subcommittee compliance test specification, the Honeywell components will be verified to inter-operate with a single third-party client and/or server (as appropriate). For each tested component, the following items will be verified to operate correctly between the Honeywell component and the third party component:

- All non-optional OPC methods
- Any optional OPC methods supported by the Honeywell component and the third party component
- All OPC-specified data types

Qualified components



REFERENCE - EXTERNAL

Refer to the "Third-Party Software Considerations" section of the SCN (Software Change Notice) for your GUS or APP release.

10 Third Party Component Planning

10.1 Third Party Component Policy

General approach for using non-tested components

Non-tested third-party components can function successfully on the TPS system; however, Honeywell has not verified the reliability of TPS nodes running the non-tested components.

However, by following the general steps outlined below, the risk of using a third-party component can be reduced:

Step	Action
1	Ensure that sufficient resources are available. Check CPU, memory and disk space. Check for peak use by sampling at multiple times during the day.
2	Ensure that enough network bandwidth is available (for example, the use of a monitoring tool is the best way to do this).
3	If SafeView is used, ensure SafeView can properly manage the component.
4	For components that use process information (for example, request LCN date), ensure that sufficient data throughput (parameters per second) to the LCN is available.
5	For hardware components, ensure that the component can be recognized, or at least does not conflict with existing standard TPS node hardware settings for IRQ and device addresses.
6	Ensure that the component installs and uninstalls properly.
7	After installation of the component, ensure that the TPS node operates correctly before starting the component testing.

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